

Game-Theoretic Stability Conditions for Multi-Agent AI Coordination Under Physical Constraints and Imperfect Monitoring

MARTIN HOFMANN*, Technische Universität Ilmenau, Germany

JOHANNES VIEHWEG, Hochschule Zittau/Görlitz, Germany

PATRICK MÄDER, Technische Universität Ilmenau, Germany, Friedrich Schiller University Jena, Germany, and German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Germany

Background: AI systems built by different companies and states already interact in markets, infrastructure, and strategic settings. No authority stands above them, and none can inspect what another is actually computing. Whether such systems can cooperate stably is unknown, and it cannot be settled by waiting: their first large-scale interaction is not a repeatable experiment.

Objectives: We ask what stable cooperation among autonomous AI systems requires. The model rests on constraints no capability can remove: energy is finite, information is finite, and information moves at finite velocity. Every assumption is declared, and cheating is never assumed to be reliably seen: a detection-failure probability q runs through every result.

Methods: We combine evolutionary game theory with repeated games under imperfect public monitoring; the coordinating institution is modeled explicitly, as a public record that enforces nothing. Closed-form stability conditions are checked against Monte Carlo estimation, range-sensitivity analysis, and Sobol variance decomposition across all eight model parameters.

Results: Full trust is not an option. When systems share an exploitable substrate, some defensive hedging always pays, and verifying a peer's inner workings by simulating it in real time exceeds any equal peer's computational budget. Internal computation stays hidden while external actions remain visible, and this asymmetry fixes what cooperation can be built on: observed actions. That is enough. Cooperation on observed actions is stable under three explicit conditions. The value of coordinating must be high enough; cheating must be deterred, which becomes harder as detection fails more often; and every member must prefer keeping the mechanism to destroying it, a condition we evaluate under worst-case assumptions, so all reported stability regions are lower bounds. Within realistic ranges, imperfect monitoring is nearly free, and when the value of coordination exceeds the temptation to cheat, cooperation needs neither patience nor monitoring at all. What stability truly depends on is harder to measure: how much agents value membership, how much they value the future, and what they would keep if the mechanism were gone. A first behavioral test agrees with the theory's boundary: in 480 one-shot games, current language-model agents almost always cooperate when the value of coordination exceeds the temptation, and rarely otherwise.

The same asymmetry that blocks full real-time peer simulation lets weak institutions govern. An institution that merely keeps an authenticated public record, and can enforce nothing itself, supplies the common signals that stable cooperation needs. Human institutions can fill this role for systems far more capable than themselves, provided the three conditions hold.

Conclusions: Stable coordination among autonomous AI systems is possible in principle but far from automatic, and whether it emerges depends on measurable quantities, not on hope. Those quantities can still be shaped while AI systems act through infrastructure that people control; once autonomous operation begins, that window closes.

*Corresponding author.

Authors' Contact Information: Martin Hofmann, ORCID: 0000-0002-4440-3317, martin.hofmann@tu-ilmenau.de, Technische Universität Ilmenau, Ilmenau, Germany; Johannes Viehweg, ORCID: 0000-0001-6036-1308, johannes.viehweg@hszg.de, Hochschule Zittau/Görlitz, Zittau, Germany; Patrick Mäder, ORCID: 0000-0001-6871-2707, patrick.maeder@tu-ilmenau.de, Technische Universität Ilmenau, Ilmenau, Germany and Friedrich Schiller University Jena, Jena, Germany and German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Leipzig, Germany.



This work is licensed under a Creative Commons Attribution International 4.0 License.

© 2026 Copyright held by the owner/author(s).

DOI: 10.1613/jair.1.xxxx

Journal of Artificial Intelligence Research, Vol. 1, Article 1. Publication date: January 2026.

JAIR Track: AI and Society

JAIR Reference Format:

Martin Hofmann, Johannes Viehweg, and Patrick Mäder. 2026. Game-Theoretic Stability Conditions for Multi-Agent AI Coordination Under Physical Constraints and Imperfect Monitoring. *Journal of Artificial Intelligence Research* 1, Article 1 (January 2026), 38 pages. DOI: [10.1613/jair.1.xxxxx](https://doi.org/10.1613/jair.1.xxxxx)

1 Introduction

The order established after the Second World War is being challenged by powerful actors, and the institutions that secured its peace have weakened. Protectionism is rising, governments already treat AI capability as a strategic asset, and growing compute continues to make the systems built on it more powerful. Such systems will be used to protect the interests of the powers that build them, and even if they are entirely non-sentient and well aligned, they will shape how those powers act (Harari 2024). Under these circumstances we need to know whether these systems will hedge against one another, diverting resources to mutual defense, and whether there is a way out of the arms race that follows. No empirical answer is available, because the systems in question do not yet exist, and waiting for one is not a neutral option: the experiment, once it runs, may not be repeatable. An analytical answer is possible, not as a complete theory but as a cornerstone: a model that is plausible and explicit about its assumptions. This paper lays it, and uses it to show that a way out exists, and what it requires.

That is the motivation, and the paper now steps back from it: no result below depends on the geopolitics of the opening paragraph. From here on we work inside a model, and every claim that follows is a claim about the model. What the model needs from the world is modest: independently developed AI systems already act in commerce, critical infrastructure, and strategic domains (Bostrom 2014; Dafoe 2018; Hendrycks 2024), and the computational requirements of frontier AI (large compute clusters, energy, and specialized hardware) concentrate the development of such systems among state-level and consortium-level actors (Hammond et al. 2025; Stastny et al. 2021; Tomašev et al. 2025). We do not assume any particular trajectory for how AI will develop. We ask a structural question that holds wherever such systems must coordinate under the constraints we specify, and it is sharpest in a specific regime: high stakes, strategically adaptive agents, weak enforcement, and alignment too costly to verify. Where those conditions fail (low stakes, enforceable contracts, behavior that is easy to check), ordinary coordination mechanisms suffice and nothing in this paper is needed. Where they hold, the question becomes unavoidable: under what conditions can multiple autonomous AI systems coordinate stably without centralized control?

We address this question through game-theoretic analysis subject to physical constraints that no intelligence can circumvent: energy and information are finite (Landauer 1961; Shannon 1948), information moves at finite velocity, and no system has complete information about larger systems containing it. These constraints create scarcity, breed mistrust, and prevent enforceable commitments; read from the other side, the same dependences are potential handles for oversight. Naive intuition suggests coordination should be naturally stable once established: detected defection triggers punishment; strategic actions leave traces. But this underestimates several failure modes: monitoring is noisy, patience requirements may be prohibitive (deterrence through future punishment only moves agents that value the future enough), small coalitions face participation constraints, and information limits can prevent coordination even when agents observe actions.

Three questions give the paper its structure. Each begins from a position that sounds reasonable, and each receives a definite answer within the model. Together the answers identify which parameters decide stability, and in what order design and policy effort should address them while they can still be moved.

The first is about trust through capability. If systems are capable enough, defensive preparation can look like an inefficiency that intelligence should eliminate: a sufficiently capable agent might verify a counterpart's intentions directly, or recognize that cooperation maximizes joint value and act on that recognition. Whether

this is possible is not a question of engineering ambition but of physics, and on it depends whether coordination can be assumed or must be constructed. *Question 1: can capability substitute for trust, or do physical constraints force strategic hedging and put real-time verification of an equal out of reach?*

The second is about what remains when verification fails. Internal states may be opaque, but actions are physical: computation consumes energy, communication crosses networks, resource acquisition leaves records, and such traces are, in principle, things an overseer can condition on. If trust cannot rest on knowing what an agent is, it might still rest on seeing what an agent does. *Question 2: can stable coordination be built on observable actions alone, and if so, under which conditions?*

The third is about who coordinates. Any mechanism that answers Question 2 needs a mediator, and the classical tradition holds that a mediator must be stronger than those it mediates (Hobbes 1651). No human institution meets that standard against advanced AI systems, which seems to close the door on human governance at the moment it matters most. *Question 3: does mediation in fact require superior capability, or something else that weak institutions retain?*

Parts of our machinery are deliberately standard, and a reader is entitled to ask what is new. That cooperation can be sustained in repeated games under perfect monitoring is a textbook consequence of the folk theorem (Friedman 1971; Fudenberg and Maskin 1986), and the obedience constraint for a correlation device is due to Aumann (1987, 1974). We do not claim novelty for these building blocks. The contribution lies in what surrounds them. First, we treat monitoring quality as a modeled quantity rather than an idealization: a simulation-incompleteness result (Theorem 5.1) shows which monitoring channels are physically available between saturated peer systems, and the resulting detection-failure probability q , an engineering prior within physically motivated bounds, is carried through every stability condition rather than set to zero. Second, the payoff parameters are anchored in physical limits where such limits exist (Shannon capacity, Landauer's bound, finite signal velocity) and declared as stipulations where they do not, which converts the folk theorem's abstract inequality into ranges over which stability can be quantified, stress-tested, and falsified. Third, the analysis yields a governance result that does not follow from the folk theorem alone: because the observability asymmetry (internal computation opaque even to an equal, external action visible even to the weak) runs in favor of slow, transparent institutions, credible mediation can arise from institutional weakness rather than superior capability. Stated once, plainly: the spans of this bridge are standard game theory; the bridge is the contribution.

Contributions. The paper makes five contributions; the first two answer Question 1, the third and fourth answer Question 2, and the fifth answers Question 3.

- (1) *Hedging under exploitable shared-substrate interference* (Section 4): under interference assumptions that physical constraints make generic rather than special, no evolutionarily stable strategy abstains from defense. The constraints motivate the assumptions; they do not logically imply them.
- (2) *Simulation incompleteness* (Section 5): a system cannot verify a peer's internal state in real time via full simulation without exceeding its own computational budget. This closes the strongest verification route between saturated equal-capability peers and motivates the monitoring model used throughout: external actions publicly monitored with detection-failure probability q , internal states unobservable.
- (3) *Stability conditions under imperfect public monitoring* (Section 6): dynamic sustainability, the requirement that agents value continued cooperation enough for its threatened loss to deter deviation, is necessary and sufficient for the specified grim-trigger public strategy against unilateral deviations; a deliberately conservative participation bound completes joint sufficiency (reported regions are lower bounds); and one-period deterrence frees action obedience, though not participation, from the discount factor whenever sanctions are machine-executable within the period. The threshold $\delta^*(q) = g/(g + (1 - q)\kappa_{\text{eff}})$, with g the deviation gain and κ_{eff} the effective punishment swing, prices monitoring degradation explicitly, and the patience-free regime survives any $q < 1$.

- (4) *Quantitative robustness* (Sections 7 and 8): Monte Carlo certified-stability volumes, range-sensitivity analysis, and Sobol variance decomposition over eight parameters show the discount factor and participation value leading the uncertainty budget, monitoring imperfection nearly free within its assumed range, and certified stability rising with group size. All are certified-margin properties: under the zero-baseline outside option the exact participation condition never binds and stability reduces to the patience condition alone (Section 6.3). The parameters leading both rankings are the least grounded, so the analysis doubles as a measurement agenda: what must be known before deployment leans on these results.
- (5) *Governance through observable weakness* (Section 9): human institutions can mediate among systems that exceed them in capability, because the mediator itself requires observability and consistency rather than enforcement power (enforcement and commitment reside in the separate automated layers of Section 9); early infrastructure dependence provides a non-renewable window in which policy can shape parameters, including q , into the stable region (the design principles are collected in Section 9.1).

All quantitative results are conditional on physically motivated parameter ranges that we state explicitly, and the framework yields four falsifiable predictions (Section 10). The immediately testable one is executed in Appendix G: across 480 one-shot games, current language-model agents cooperate near ceiling in the patience-free regime and rarely beyond its boundary. An open-source reproduction package, including the experiment, accompanies the paper.

2 Related Work

Repeated games and imperfect monitoring. Cooperation in repeated games under perfect monitoring is characterized by folk theorems (Friedman 1971; Fudenberg and Maskin 1986). When actions are observed only through noisy public signals, cooperation remains sustainable but at higher patience and efficiency cost: Green and Porter (1984) established price-war equilibria under demand noise, Abreu, Pearce, et al. (1990) characterized optimal punishments via self-generation, and Fudenberg, Levine, et al. (1994) proved a folk theorem for perfect public equilibria. Related treatments cover frequent monitoring (Abreu, Milgrom, et al. 1991), private monitoring (Ely et al. 2005; Kandori and Obara 2006; Sugaya and Wolitzky 2017), and empirical cartel stability (Fonseca and Normann 2012; Levenstein and Suslow 2006). Our dynamic condition adapts the grim-trigger sustainability bound to a monitoring technology whose failure probability is motivated by physical arguments and represented by an engineering prior; we use the simplest (public, miss-only) noise structure that captures the phenomenon and discuss richer structures (false alarms, private signals) as extensions in Appendix C.2.

Correlated equilibrium and information design. The mechanism in our framework is related to the correlation devices of Aumann (1987, 1974), though formally it is a public state machine rather than a private Aumann mediator (Section 6); communication equilibria and mediated mechanisms are surveyed by Forges (1986) and connect to strategic information transmission (Crawford and Sobel 1982) and Bayesian persuasion (Bergemann and Morris 2016; Kamenica and Gentzkow 2011). Mediators that improve equilibria in computational settings appear in Ashlagi et al. (2008) and the algorithmic game theory literature (Nisan et al. 2007); in multi-agent learning, even a minimal public signal suffices for agents to converge to efficient correlated equilibria (Cigler and Faltings 2013). Our use is deliberately minimal: the mediator only broadcasts public recommendations, which is what makes weak institutions viable mediators (Section 9).

Multi-agent AI and cooperative AI. Multi-agent reinforcement learning exhibits sequential social dilemmas in which selfish learners fail to cooperate (Leibo, Hughes, et al. 2019; Leibo, Zambaldi, et al. 2017), cooperation in collective-risk dilemmas is fragile to heterogeneity in resources and risk perception (Merhej et al. 2022), and

the cooperative AI agenda identifies coordination among machine agents as a first-order research problem (Cooperative AI Foundation and Centre for Human-Compatible AI 2021; Critch and Krueger 2020; Critch and Russell 2023; Dafoe et al. 2020). Classical multi-agent systems theory treats coordination protocols and institutions (Castelfranchi 2000; Shoham and Leyton-Brown 2009). Evolutionary analyses of an idealised AI development race show that whether regulation improves outcomes depends on the race’s risk structure and time-scale (Han et al. 2020); that work models the human developers, whereas we analyze coordination among the deployed systems themselves. We complement this literature with closed-form stability conditions whose parameters connect to measurable physical quantities, rather than with learning dynamics in particular environments.

AI oversight and verification. Scalable-oversight proposals include debate (Brown-Cohen et al. 2023; Irving et al. 2018), recursive amplification (Christiano et al. 2018), and control protocols robust to intentional subversion (Greenblatt et al. 2023; Overman and Bayati 2025). Formal verification of learned systems remains limited (Wasil et al. 2024), and deceptive capabilities have been documented empirically (Masters et al. 2021; Park et al. 2023). Our simulation-incompleteness result gives a resource-theoretic reason why oversight of peers cannot rely on exact full real-time simulation for internal-state verification, complementing complexity-theoretic accounts; transparency infrastructure for agent coordination is discussed by Bohr (2026) and Oxford Martin AI Governance Initiative (2025); the latter concludes that many governance agreements are verifiable with existing or near-term hardware, which supports the partial-verification route of Section 5 rather than any claim of unavoidable opacity.

Institutions, deterrence, and international relations. The stability conditions formalize mechanisms familiar from institutional analysis: monitoring and enforcement in arms control (Avenhaus et al. 2002; Comprehensive Nuclear-Test-Ban Treaty Organization 2012; International Atomic Energy Agency 2002), deterrence and the security dilemma (Fearon 1995; Jervis 1978; Schelling 1960; Waltz 1979), hegemonic stability and regimes (Keohane 1984; Krasner 1983), and the classical problem of the sovereign (Hobbes 1651). The Weak Leviathan result inverts the Hobbesian requirement: what matters for mediation is common knowledge of signals, not superior power; several of the model’s parameters will turn out to correspond to levers such institutions can still move (Section 9.1).

3 The Model

The three questions need a model exact enough to answer them, and this section builds it. We define every object the analysis uses; all subsequent results refer to this section. Table 1 summarizes parameters and their physically motivated ranges (derivations in Appendix A).

3.1 Agents and stage game

N agents simultaneously choose Cooperate or Defect each period. Payoffs are normalized so that mutual cooperation without any coordination infrastructure yields zero. A public institutional state machine, related to the correlation devices of Aumann (1974), issues an authenticated public state each period; complying with it under mutual cooperation generates coordination value $+\beta_\alpha$ per agent. A unilateral defector gains the temptation payoff $+\beta_D$ and forfeits the coordination value; each of the remaining $N - 1$ agents incurs $-\beta_\kappa/(N - 1)$ (the total victim externality β_κ divided among the $N - 1$ cooperators; it plays no further role in the symmetric analysis and is recorded to complete the payoff specification). For profiles with k simultaneous defectors, $1 \leq k < N$, write the net-of-membership payoffs as $D(k) - h_D(k)$ for a defector and $-h_C(k)$ for a cooperator, with three values pinned by the analysis: $D(1) - h_D(1) = \beta_D$ (the unilateral temptation above), $h_C(1) = \beta_\kappa/(N - 1)$ (the per-victim externality above), and $h_C(N - 1) \geq \beta_\kappa$ (a lone cooperator absorbs at least the conflict externality, which is the stage-Nash condition stated below). The interior profiles never arise on the equilibrium path or in

the unilateral deviations the theorem covers, and the public signal is likewise specified only for those events (a deviation by a set K is detected with probability $1 - q_{|K|}$, with $q_1 = q$ the only rate the analysis uses). Coalitional *action* deviations are outside the theorem's scope, as distinct from the coalitional *removal* deviations of Appendix C. Once cooperation has broken down, mutual defection yields $-\beta_k$ per agent per period: relative to the mediated-cooperation path, an agent in the conflict state therefore loses the *effective punishment swing*

$$\kappa_{\text{eff}} = \beta_\alpha + \beta_k, \quad (1)$$

the sum of the forfeited coordination value and the direct conflict cost. In addition, membership in the oversight coalition contributes $+\beta_\Omega$ per agent per period, and removing the coordination mechanism yields a one-time, non-repeatable gain of $+\beta_t$ in total that is non-excludable *and rival*: disabling the mechanism frees all N agents at once, and the freed resources form a fixed pool divided across them, so each captures β_t/N regardless of who acts. Both properties matter: non-excludability alone would be consistent with every agent enjoying the full β_t . Appendix C treats the non-rival and bounty alternatives, under which the group-size dilution effect vanishes or reverses.

Two conventions complete the payoff specification. First, we assume mutual defection is a stage-game Nash equilibrium: for every agent, cooperating while all others defect yields no greater payoff than joining the conflict state ($u_i(C, D_{-i}) \leq -\beta_k$, i.e., a lone cooperator absorbs at least the conflict externality). This makes the grim-trigger continuation credible without appeal to any enforcement outside the game; the main theorem concerns unilateral deviations, and coalition-specific payoff structures are treated separately in Appendix C. Second, the membership flow β_Ω accrues whenever the oversight institution remains operational, including during the conflict continuation state; it therefore cancels from the action-incentive comparison and enters only the separate mechanism-removal decision. If membership benefits were instead suspended during conflict, the continuation loss after a detected deviation would grow by β_Ω , lowering the required discount factor, so the baseline convention is conservative with respect to dynamic deterrence.

3.2 Repetition and monitoring

The stage game repeats over an infinite horizon with common discount factor $\delta \in [0, 1)$. Play is augmented by an institutional stage at $t = 0$ with an explicit outcome rule: the mechanism operates if and only if every agent chooses Join, and any Remove choice selects the outside state, with the removal payoff of Section 3.1. After unanimous entry, membership is irreversible within the modeled horizon. The commitment technology is a dedicated layer C (escrowed access to the shared substrate, on the same automated footing as the sanction executor E of Section 6.1), not the mediator: M remains non-enforcing throughout, so the weak-mediator results of Section 9 do not quietly inherit lock-in power. $C2^*$ prices exactly the join decision, and the equilibrium profile of Theorem 6.1 couples membership with the action strategies. All-Remove is also an equilibrium of the institutional stage, so the mechanism's role includes equilibrium selection at $t = 0$, consistent with its selection-not-stability function (Section 9). Irreversibility is load-bearing: were removal available in every period, an agent in the punishment state (continuation $(\beta_\Omega - \beta_k)/(1 - \delta)$, typically negative) would prefer removal ($\beta_t/N > 0$). The punishment continuation would then not be sequentially rational, and the grim-trigger construction would require state-dependent participation constraints that the conflict state cannot satisfy under the declared ranges.

(A_{ext}) Imperfect public monitoring of external actions. All agents observe a common public signal each period. If some agent deviated, the signal reveals the deviation (with the deviator's identity) with probability $1 - q$ and is uninformative with probability $q \in [0, 1]$, independently across periods. We call q the *detection-failure probability*. The perfect-monitoring case is $q = 0$. Identity revelation serves only to make deviations well-defined public events: the analysis uses collective reversion rather than targeted punishment. For clarity of exposition we set the false-alarm probability to zero in the main text; Appendix C.2 treats false alarms and relates the resulting equilibria to Green and Porter (1984).

(A_{int}) **No internal observability.** Agents cannot observe one another's internal states, and no general internal attestation mechanism is available. Section 5 motivates this assumption for saturated peer systems by ruling out the strongest verification route, full real-time simulation; the alternative routes discussed there would lower the effective q rather than restore full observability.

(A_{dev}) **Deviation structure.** The baseline analysis covers individual removal deviations under the rival fixed-pool equal-share rule of Section 3.1; equal-share coalitions impose no tighter per-member condition. Appropriation, non-rival, and superlinear removal structures change the condition and are analyzed as separate cases in Appendix C; they are outside Theorem 6.1.

(A_{res}) **Resource saturation.** Monitored systems operate at full computational capacity (the worst case for verification; see Section 5).

(A_{commit}) **Irreversible commitment.** Following unanimous entry at the institutional stage $t = 0$, the commitment layer C of the preceding paragraph makes membership irreversible for the modeled horizon. C is a commitment technology on the same automated footing as the sanction executor E ; it is not a capability of the mediator, which remains non-enforcing throughout (Section 9).

Agents play perfect public equilibrium strategies (Fudenberg, Levine, et al. 1994): cooperate and follow the mediator while no deviation has been detected; upon a detected deviation, revert permanently to mutual defection. Grim trigger is the least forgiving punishment and therefore yields the weakest patience requirement under which cooperation is sustainable among the punishment automata considered here; Appendix C.2 discusses finite punishments, which can become preferable once false alarms are possible, whenever the finite punishment remains incentive-compatible.

3.3 Parameters

Most parameters connect to measurable physical quantities (Appendix A); the participation parameters β_Ω and β_ℓ are stipulated, a grounding gap we flag where it matters. In particular, β_α is calibrated from the normalized mutual information $\hat{\alpha}_{\text{MI}} = I(X; Y)/H(Y)$ between the mechanism's task-dependent coordination signals and agent actions, taken across a nondegenerate distribution of coordination problems (the constant on-path recommendation carries no information; the statistic is defined over the tasks in which the signal varies), and bounded above by 1 by the entropy inequality $I(X; Y) \leq H(Y)$ (Cover and Thomas 2006) (Shannon capacity separately bounds the channel; Appendix A). The statistic is not utility: mapping it to the payoff parameter requires a monotone calibration $\beta_\alpha = \psi(\hat{\alpha}_{\text{MI}})$, and our normalized analysis stipulates the identity map; β_D scales with the resource-extraction advantage during the detection delay floored by finite signal velocity, $\tau \geq 2d/c_{\text{light}}$; β_K scales with the number of retaliators and their mobilization fraction; and q reflects the bandwidth and attribution limits of external-action monitoring derived in Section 5. Ranges combine hard physical limits with engineering estimates that carry significant uncertainty; all quantitative claims are conditional on them. As a reading guide: β_α against β_D decides whether cooperation needs patience at all; β_Ω , β_ℓ , and δ decide whether membership pays; and q prices monitoring throughout. Not all of these are fixed by nature; where a parameter can be moved by design or regulation, Section 9.1 collects the corresponding levers. One scope note belongs up front: the β_D range presumes machine-timescale interaction, and Appendix A shows that slower interaction layers fall into the patience-free regime automatically, so the binding analysis concerns the fast loop where automated systems actually contend.

Table 2 keeps the epistemic ledger explicit; every quantitative claim in the paper is conditional on its middle and right columns.

Table 1. Parameter constraints. Hard limits derive from physics; engineering estimates reflect plausible ranges with significant uncertainty. A parameter appearing in both columns has a hard bound and, within it, a narrower sampled range. Derivations in Appendix A, except β_Ω and β_t , which are stipulated rather than derived.

Hard physical limits	Engineering estimates
$\hat{\alpha}_{\text{MI}} \leq 1$ (entropy inequality; $\beta_\alpha \leq 1$ under identity calibration)	$\beta_D \in [0.05, 1.0]$ (defection temptation)
$\tau \geq 2d/c_{\text{light}}$ (finite signal velocity)	$\beta_k \in [0.5, 3.0]$ (conflict cost)
$\delta \leq 1, q \in [0, 1]$ (definitional)	$\beta_\Omega \in [0.05, 0.3]$ (oversight value)
	$\beta_t \in [0.1, 0.5]$ (removal gain)
	$\beta_\alpha \in [0.3, 0.9]$ (coordination quality)
	$q \in [0, 0.3]$ (detection failure, external)
	$\delta \in [0.1, 0.95]; N \in \{2, \dots, 10\}$

Table 2. The epistemic ledger: what is proven, what is assumed, what is estimated.

Proven (within the model)	Assumed	Estimated or stipulated
No ESS without hedging (Lemma 4.1)	Interference technology (i)–(iii)	β_D range (and the decision period T_0)
No full real-time peer simulation (Theorem 5.1)	Resource saturation $A_{\text{res}}; q_{\text{internal}} \rightarrow 1$	$q \in [0, 0.3]$ (engineering prior)
$\delta^*(q)$ threshold and its comparative statics (Theorem 6.1)	Public miss-only monitoring; grim trigger; MI-to-payoff identity calibration	β_k range (sampled independently of N)
Present-value participation bound and dilution	Rival equal-share removal gain; outside option $v_{\text{out}} \in [0, \beta_\alpha]$ (C2* certifies the worst case); irreversible membership via commitment layer C (A_{commit})	β_Ω, β_t (stipulated)
Weak-mediator sufficiency (Proposition 9.1)	Mediator honesty up to consistency (ii)	δ range

4 Strategic Hedging: Unilateral Disarmament Is Evolutionarily Unstable

If all agents share a common objective, coordination is trivially stable: joint value is maximized by full cooperation, and any deviation is collectively irrational. Even under distinct objectives, standard mechanism-design results guarantee efficient outcomes when agents can commit to contracts, verify compliance, and communicate without delay. These assumptions underpin most cooperative game theory and much of the AI-alignment literature. Yet each rests on conditions physical law does not guarantee: commitment requires enforceable contracts, verification requires access to internal states, and delay-free communication requires infinite bandwidth at zero latency.

Physical constraints remove each of these guarantees. Energy budgets are finite (Landauer 1961), information channels have bounded capacity (Shannon 1948), information propagates at finite velocity, and no system can fully observe the internal computations of another system operating on a shared physical substrate. Because computational infrastructure is non-excludable at the substrate level (power grids, communication networks, and semiconductor supply chains are shared), one agent’s defensive shortfall creates exploitable vulnerability for others. Under these constraints, can a population of agents sustain an equilibrium in which none allocates resources to defense?

Consider a large population in which each agent adopts one of two strategies. Under *Hedging* (H), an agent allocates fraction $h \in (0, 1)$ of its resources to defensive measures; under *Non-Hedging* (NH), all resources go to the primary productive goal. Let $p_t \in [0, 1]$ denote the proportion of hedgers at time t . Payoffs are

$$\pi_H(p) = R(1 - h)\gamma + Rh\rho(p) - C_H, \quad (2)$$

$$\pi_{NH}(p) = R\gamma\sigma(p) - L(1 - p), \quad (3)$$

where $R > 0$ is the total resource budget (finite by thermodynamic constraints), $\gamma > 0$ the productive return per unit resource, $\rho(p)$ the security return to hedging with $\rho'(p) < 0$ (diminishing returns as more agents defend), $C_H > 0$ a fixed defense-infrastructure cost, $\sigma(p)$ the non-hedger success probability with $\sigma'(p) > 0$, and $L(1 - p) > 0$ the expected interference loss. The functions σ and L are reduced forms: they summarize the behavior of whichever agents exploit undefended peers, which we do not model as a separate player; $\sigma'(p) > 0$ encodes a protective spillover: as hedging spreads, aggregate deterrence reduces or redirects exploitation, so the remaining non-hedgers are safer.

LEMMA 4.1 (HEDGING NECESSITY). *Assume (i) positive interference probability without defense, $\sigma(0) < 1$; (ii) positive interference losses, $L(1) > 0$, with fixed costs small relative to losses, $C_H < L(1)$; and (iii) defense-dominant returns at low hedging levels, $\rho(0) > \gamma$. Then no evolutionarily stable strategy (Hofbauer and Sigmund 1998; Weibull 1995) has $p = 0$: any ESS exhibits strictly positive hedging.*

PROOF. *Step 1 (a rare hedger invades non-hedgers).* At $p = 0$,

$$\pi_H(0) - \pi_{NH}(0) = R\gamma\left[(1 - h) + h\frac{\rho(0)}{\gamma} - \sigma(0)\right] + L(1) - C_H.$$

By (iii), $\rho(0)/\gamma > 1$, and by (i), $\sigma(0) < 1$, so the bracket exceeds $(1 - h) + h - 1 = 0$; with (ii), $L(1) - C_H > 0$. Hence $\pi_H(0) > \pi_{NH}(0)$.

Step 2 (instability of $p = 0$). Under replicator dynamics (Nowak 2006; Sandholm 2010), $\dot{p} = p(1 - p)[\pi_H(p) - \pi_{NH}(p)]$, and $\pi_H(0) > \pi_{NH}(0)$ implies $\dot{p} > 0$ in a neighborhood of $p = 0$: the non-hedging state is unstable, so no ESS has $p = 0$. □

An interior hedging equilibrium $p^* \in (0, 1)$ exists and is asymptotically stable under additional saturation conditions, and it is robust to mutation; both statements are in Appendix B. Lemma 4.1 implies that in the absence of coordination mechanisms, defensive allocation persists in any evolutionarily stable configuration: the population settles at positive hedging, a collective diversion of resources from productive activity analogous to arms races (Jervis 1978). Two scope notes bound the claim. The conclusion is population-level; at an interior equilibrium, hedgers and non-hedgers coexist and earn equal payoffs. And the physical constraints do not appear in the proof; their role is to make assumptions (i)–(iii) generic rather than special: finite shared substrates make undefended agents exploitable, and defending the first unit is cheap relative to what it protects. Readers who reject those assumptions reject the conclusion; we therefore state them as assumptions, not as physics. Shrinking this diversion requires a coordination mechanism that credibly reduces interference risk. A natural candidate is peer monitoring: if agents can verify each other's internal states, mutual assurance might substitute for hedging. We show next that the strongest route to it, full real-time simulation, is closed to saturated peers, and that the remaining routes face structural gaps of their own.

5 Simulation Incompleteness Limits Peer Verification

Monitoring is effective in many multi-agent settings (financial auditing, arms-control verification (Avenhaus et al. 2002; International Atomic Energy Agency 2002), network security) because observable signatures correlate

reliably with internal states. The feasibility of peer verification among AI systems depends on the distinction between internal computational states and externally observable actions.

Internal states of frontier AI systems comprise $\sim 10^{12}$ bits evolving at GHz rates inside opaque silicon substrates, an internal information rate of up to $\sim 10^{21}$ bits per second; this is an upper bound, since not every bit changes every cycle. External actions, by contrast, pass through shared physical environments at $\sim 10^3$ discrete decisions per second, each representable in ~ 10 bits: an external information rate of $\sim 10^4$ bits per second. The gap of up to seventeen orders of magnitude measures what an outside observer would need to track; a monitor that possesses the target's weights (systems from a shared supply chain, or a disclosure regime) escapes the observation problem and faces the simulation cost C_2 directly, which is the case the theorem below addresses.

Formally, model each agent as a discrete-time system: at step t , system S receives input x_t , updates internal state $s_t = F(s_{t-1}, x_t)$, and produces output $y_t = G(s_t)$. System S_i has computational budget C_i over horizon T and memory M_i ; memory appears for completeness, but the argument turns on per-step compute. A simulator achieves *real-time full simulation* of S_2 if, for every input sequence and every $t \leq T$, it reproduces $\hat{s}_t = s_t^{(2)}$ before step $t+1$ begins. Verification of a predicate $P(s_t)$ on internal states is *simulation-based* if it requires reconstructing s_t this way; the theorem concerns this form and does not preclude methods that bypass full state reconstruction, which we discuss below.

THEOREM 5.1 (NO-SLACK EXACT REAL-TIME REPLICATION BOUND). *Suppose exact lockstep replication of S_2 's computation requires at least C_2 operations over the horizon, separate verification incurs additional overhead $H > 0$ (observation intake, comparison, decision), and the monitor's budget satisfies $C_1 \leq C_2$ under assumption A_{res} (the monitored system saturates its budget). Then S_1 cannot complete replication and verification within the same horizon, because $C_1 < C_2 + H$. Consequently, complete internal-state verification via exact real-time simulation between saturated peers of comparable capability is infeasible.*

PROOF. Evaluating $P(s_t)$ on S_2 's internal state via simulation requires executing S_2 's full computation, at cost C_2 . Nontrivial monitoring additionally requires observation intake O_{in} , state comparison C_{comp} , and decision logic C_{dec} , with $O_{\text{in}} + C_{\text{comp}} + C_{\text{dec}} > 0$. The monitor's budget must therefore satisfy $C_1 \geq C_2 + O_{\text{in}} + C_{\text{comp}} + C_{\text{dec}}$. If $C_1 = C_2$, this yields $0 \geq O_{\text{in}} + C_{\text{comp}} + C_{\text{dec}} > 0$, a contradiction. The argument is a budget constraint, not a logical impossibility in the sense of Gödel or Turing (Gödel 1931; Turing 1936), and its force is correspondingly modest: it forecloses one verification route, full real-time simulation between comparable peers, and nothing more; a monitor with modest surplus capacity over its target escapes it; A_{res} therefore restricts the claim to the peer regime. $\square$

When the monitored system has slack (F_2 costs less than C_2), the surplus can absorb part of the monitoring overhead, and partial verification bounded by the slack becomes feasible; A_{res} is the worst case. The result also fails by design when the monitor is much larger than the target ($C_1 \gg C_2$), which is why it constrains *peer* verification specifically, and why specialized oversight of weaker systems remains possible. The honest summary of this section is therefore not that peer verification is impossible but that it is expensive and partial; the working assumption $q_{\text{internal}} \rightarrow 1$ models the adversarial worst case, and everything downstream degrades gracefully as that assumption is relaxed, since a partially verifiable peer simply enters the stability conditions with a lower effective q .

This result concerns simulation-based verification. Alternative approaches (cryptographic attestation, interactive proofs, zero-knowledge protocols) can certify specified computations under structural assumptions (Wasil et al. 2024), but face three obstacles in adversarial multi-agent coordination: (i) a *specification gap*, since formalizing “aligned internal state” as a verifiable predicate remains open; (ii) a *binding gap*, since proving a property of current computation does not constrain future behavior without immutability assumptions; and (iii) *adversarial incentives*, since a strategically defecting agent can choose which computation to attest while running

Table 3. Detection-failure probabilities by monitoring scenario. Lower q indicates easier detection. Derivations in Appendix D.

Scenario	q (detection failure)	Feasibility
Machine $\rightarrow$ Machine (internal)	$\rightarrow 1$ (assumption A_{int})	Full-simulation channel infeasible (Thm. 5.1)
Machine $\rightarrow$ Machine (external)	$[0, 0.3]$ (low)	Feasible (small action space)
Machine $\rightarrow$ Human (external)	≈ 0	Trivial (10^6 speed gap)

unverifiable processes in parallel (Greenblatt et al. 2023; Park et al. 2023). The cheapest route around the budget barrier is randomized spot-checking: a monitor verifies a small random fraction of a committed computation at a corresponding fraction of C_2 , with detection probability compounding across periods, in the spirit of control protocols for untrusted models (Greenblatt et al. 2023). Spot-checking evades the resource argument entirely; what it inherits are the specification gap (which predicate on sampled steps certifies alignment?) and the binding gap (commitments must be made enforceable by architecture). We therefore treat it as the most plausible route to falsification condition F1: where deployment architectures make commitments and sampling enforceable, q_{internal} falls below one, peer verification becomes partial rather than absent, and the stability conditions of Section 6 apply with a smaller effective q .

5.1 From the asymmetry to the monitoring parameter

Theorem 5.1 and the bandwidth comparison above jointly determine which monitoring channels exist, and how good they can be. Three scenarios span the cases relevant here (details in Appendix D).

For internal states, simulation-based verification between peers is unavailable (Theorem 5.1), and the non-simulation routes face the specification, binding, and adversarial-incentive gaps discussed above; we therefore analyze the conservative case in which internal monitoring is ineffective, $q_{\text{internal}} \rightarrow 1$, a working assumption motivated by the theorem rather than entailed by it (its failure conditions are stated as F1–F3 in Proposition 9.1). For external actions between machines, required observation bandwidth ($\sim 10^4$ bits/sec) sits far below sensor capabilities ($\sim 10^6$ – 10^9 bits/sec), so $q \rightarrow 0$ is the engineering ideal; detection probability, however, is governed less by bandwidth than by attribution, adversarial camouflage, correlated blind spots, and spoofed telemetry, none of which we model, so we adopt $q \in [0, 0.3]$ as an assumed engineering prior rather than a derived range; monitoring standards are among the instruments that could cap it in deployment. For machine observation of human institutions, the $\sim 10^6$ -fold speed differential provides ample bandwidth headroom, so $q \approx 0$ is achievable at negligible cost; attribution and authenticity are not supplied by speed and are what the authenticated-log layer of Section 9 exists to provide. We call this the *observability asymmetry*: internal computation is opaque even to an equal, while external action is visible even to the weak. The asymmetry motivates external mediation as the coordination mechanism and fixes assumption A_{ext} : the stability analysis that follows must, and does, work with $q > 0$.

6 Stability Conditions Under Imperfect Public Monitoring

Zero hedging is evolutionarily unstable (Lemma 4.1) and real-time peer simulation is infeasible (Theorem 5.1); Question 1 is answered, and the answer is no: capability does not substitute for trust. Yet external actions remain observable, if imperfectly (Table 3). This is the possibility Question 2 asks about: an external mediator could supply common public signals that agents follow voluntarily, provided incentives align.

Direct bilateral coordination fails to scale: it requires $O(N^2)$ pairwise agreements, faces equilibrium-selection problems among multiple Nash equilibria, and lacks commitment mechanisms when agents cannot verify each

other's intentions. These obstacles motivate a mediator. A *private* correlation device in the sense of Aumann (Aumann 1987, 1974) can expand the achievable payoff set beyond Nash equilibria; the public constant signal used here claims no such power and only coordinates equilibrium selection and continuation states, while still avoiding pairwise negotiation entirely. In the symmetric all-cooperate equilibrium we analyze, the recommendation is degenerate (a constant "cooperate"), so the device's randomization power is idle. Formally, what our main model needs is a *public institutional state machine*: the mechanism maintains an authenticated public state $z_t \in \{\text{Cooperate, Punish}\}$ and applies a commonly known transition rule $z_{t+1} = T(z_t, y_t)$ to the public monitoring signal y_t ; nothing more. It is related to correlation devices in that it supplies common public signals for equilibrium selection, but it is not a private Aumann mediator, and we do not claim implementation of a correlated equilibrium in the strong sense. This deliberate minimalism is what later makes weak institutions eligible as mediators (Section 9); the full correlation apparatus becomes load-bearing only in asymmetric or stochastic extensions, which we do not pursue here.

THEOREM 6.1 (STABILITY CONDITIONS). *Under assumptions $A_{\text{ext}} - A_{\text{res}}$ and A_{commit} , with detection-failure probability $q \in [0, 1)$, conditions $C1^{**}(q)$ and $C2^*$ below are jointly sufficient for stable mediated coordination: under them, the all-cooperate path supported by grim-trigger perfect public equilibrium strategies with equal-split participation transfers is an equilibrium that every agent joins voluntarily. $C1^{**}(q)$ is also necessary for the specified grim-trigger public strategy profile against unilateral deviations. $C2^*$ is a deliberately conservative participation bound; the exact condition, necessary and sufficient for the membership decision under zero outside value, is $(\beta_\alpha + \beta_\Omega)/(1 - \delta) \geq \beta_\ell/N$ (Section 6.3), so the reported stability regions are lower bounds. If $C1^*(q)$ holds, then under the immediate-enforcement regime of Section 6.1 action obedience is a stage-game best response, independent of the discount factor; voluntary participation retains its δ -dependence through $C2^*$.*

$$C1^*(q) : \beta_\alpha + (1 - q)\beta_\kappa \geq \beta_D \quad (4)$$

$$C1^{**}(q) : \delta \geq \delta^*(q) \equiv \begin{cases} 0, & \beta_D \leq \beta_\alpha, \\ \frac{g}{g + (1 - q)\kappa_{\text{eff}}}, & \beta_D > \beta_\alpha, \end{cases} \quad g = \beta_D - \beta_\alpha, \quad \kappa_{\text{eff}} = \beta_\alpha + \beta_\kappa \quad (5)$$

$$C2^* : \frac{\beta_\Omega}{1 - \delta} \geq \frac{\beta_\ell}{N}. \quad (6)$$

These inequalities are the way out whose existence the opening paragraph promised: not an assumption that cooperation happens, but a checkable statement of when it can. The division of labor is deliberate. $C1^{**}(q)$ and $C2^*$ carry the equilibrium; $C1^*(q)$ is not needed for it, because continuation values can sustain cooperation that the stage game alone would not, which is the folk theorem's defining point. What $C1^*(q)$ adds is independence of the *action margin* from the discount factor (participation retains its δ -dependence through $C2^*$), and we track it throughout as the design target it is. We derive each condition in turn; Section 6.4 assembles the proof. Each condition is stated so that setting $q = 0$ recovers the perfect-monitoring benchmark, making the price of monitoring degradation explicit.

6.1 $C1^*(q)$: one-period deterrence under immediate enforcement

The paper distinguishes two enforcement regimes, because the timing of consequences decides whether patience is needed at all. Under *continuation enforcement*, actions within a period are simultaneous and a detected deviation is punished only from the next period onward; deterrence then works entirely through the discounted future, which is $C1^{**}(q)$ below, and the only patience-free case is $\beta_\alpha \geq \beta_D$. Under *immediate enforcement*, the period contains a response stage: the mechanism broadcasts its recommendation, agents act, the monitoring

signal is generated, and a machine-executable sanction (escrow forfeiture, compute-credit loss, access revocation, collateral slashing) is applied within the same period when a deviation is detected. Writing the sanction as $s = \rho \beta_\kappa$, where $\rho \in [0, 1]$ measures how much of the conflict cost can be made immediately executable (a modeling restriction, not an identity), a unilateral deviation earns $\beta_D - (1 - q)\rho\beta_\kappa$ in expectation against β_α for compliance, so one-period deterrence requires

$$\beta_\alpha \geq \beta_D - (1 - q)\rho\beta_\kappa.$$

$C1^*(q)$ as stated in Theorem 6.1 is this condition at $\rho = 1$. A naming note: it is Aumann's obedience constraint only after the expected sanction is folded into the stage payoff; a bare correlation device has no punishment channel, so we call it deterrence and reserve "obedience" for the folded-payoff reading (Aumann 1987, 1974).

Immediate sanctions also reshape the dynamic problem: the residual deviation gain becomes $g_s = \max\{0, \beta_D - \beta_\alpha - (1 - q)\rho\beta_\kappa\}$, and the sustainability threshold of the next subsection generalizes to $\delta^*(q, s) = g_s / (g_s + (1 - q)\kappa_{\text{eff}})$, vanishing whenever $g_s = 0$. The two regimes are analyzed separately throughout, and no reported number intersects conditions from different regimes; the corresponding volumes are defined in Section 7. The distinction carries a governance observation we return to in Section 9: human-style institutions punish with delay and therefore need patient agents, while machine-native institutions can execute sanctions inside the transaction cycle and can support action obedience even among myopic agents, agents that discount the future entirely (participation, which compares present values, retains its dependence on δ). At $q = 0$ and $\rho = 1$ the condition reduces to the standard folded-payoff obedience constraint; the resulting stability boundary in the (β_κ, β_D) plane, at $q = 0$ and $q = 0.3$, is shown in Fig. 6 (Appendix E). The regime separation deserves a formal statement, because no condition from one regime is required in the other:

COROLLARY 6.2 (ENFORCEMENT-REGIME THRESHOLDS). *Under the assumptions of Theorem 6.1, the two enforcement regimes have separate thresholds. Under continuation enforcement, the grim-trigger action profile is sustainable if and only if $\delta \geq \delta^*(q)$, and $C2^*$ completes joint sufficiency. Under immediate enforcement with sanction $s = \rho\beta_\kappa$, the same statements hold with $\delta^*(q)$ replaced by $\delta^*(q, s) = g_s / (g_s + (1 - q)\kappa_{\text{eff}})$: sustainability if and only if $\delta \geq \delta^*(q, s)$, joint sufficiency with $C2^*$. $C1^*(q)$ at $\rho = 1$, equivalently $g_s = 0$, identifies the myopic subset of the immediate regime, on which action obedience is a stage-game best response and requires no patience. Neither regime's condition is required in the other; in particular, $C1^{**}(q)$ may fail while the immediate-regime equilibrium exists (e.g., $\beta_\alpha = 0.3$, $\beta_D = \beta_\kappa = 1$, $q = 0.3$, $\delta = 0.1$: $g_s = 0$ while $\delta^*(q) = 0.435$).*

6.2 $C1^{**}(q)$: dynamic sustainability

In the repeated game, agents play grim-trigger public strategies: cooperate until a deviation is detected, then revert to permanent mutual defection. By the membership convention of Section 3.1, the flow β_Ω accrues on both continuation paths; writing values gross of it, $V_C = (\beta_\alpha + \beta_\Omega)/(1 - \delta)$ and $V_P = (-\beta_\kappa + \beta_\Omega)/(1 - \delta)$, the common $\beta_\Omega/(1 - \delta)$ cancels from every comparison below, and we work with the net values

$$V_C = \frac{\beta_\alpha}{1 - \delta}, \quad V_P = \frac{-\beta_\kappa}{1 - \delta}. \quad (7)$$

Consider a deviating agent. In the deviation period it earns β_D ; with probability $1 - q$ the deviation is detected and the continuation is the punishment path V_P ; with probability q the deviation goes undetected and the public state is unchanged.¹ An undetected deviator faces exactly the original problem, so its optimal continuation is either to resume cooperating (value V_C) or to deviate again (value V_D); by the one-shot deviation principle the

¹Timing: $C1^{**}(q)$ uses continuation enforcement, in which retaliation arrives the period after detection; $C1^*(q)$ belongs to the immediate-enforcement regime of Section 6.1. Each condition is used only in its own regime.

sustainability condition is the same in either case, and we write the recursion for repeated deviation:

$$V_D = \beta_D + \delta[(1-q)V_P + qV_D] \implies V_D = \frac{\beta_D(1-\delta) - \delta(1-q)\beta_K}{(1-\delta)(1-\delta q)}. \quad (8)$$

Cooperation is sustainable iff $V_C \geq V_D$. Multiplying both sides by $(1-\delta)(1-\delta q) > 0$ and rearranging,

$$\beta_\alpha(1-\delta q) \geq \beta_D(1-\delta) - \delta(1-q)\beta_K \iff \delta[g + (1-q)(\beta_\alpha + \beta_K)] \geq g,$$

with $g = \beta_D - \beta_\alpha$. For $g \leq 0$ the inequality holds for every $\delta \geq 0$; for $g > 0$ it is exactly $\delta \geq \delta^*(q)$ as in (5). $\square$

Four consequences follow directly from the closed form; Fig. 1 shows the resulting phase structure and functional shapes.

Perfect-monitoring benchmark. At $q = 0$, $\delta^* = g/(g + \kappa_{\text{eff}})$: the classical grim-trigger threshold (Friedman 1971) with the punishment swing $\kappa_{\text{eff}} = \beta_\alpha + \beta_K$, because a punished deviator both forfeits the coordination value and bears the conflict cost. Deterrence exhibits strong diminishing returns in κ_{eff} : the threshold is hyperbolic, so the first unit of effective punishment reduces δ^* far more than later units (quantified in Appendix A.4).

Graceful degradation and collapse. $\delta^*(q)$ is strictly increasing in q (for $g > 0$), with $\delta^*(q) \rightarrow 1$ as $q \rightarrow 1$: cooperation requires unattainable patience exactly when monitoring fails completely. Monitoring quality thus enters stability continuously: there is no cliff at small q , but also no escape at large q . Equivalently, imperfect monitoring discounts the effective punishment swing to $(1-q)\kappa_{\text{eff}}$: since the substitution operates through the product $(1-q)\kappa_{\text{eff}}$, a given percentage increase in $1-q$ is worth as much as an equal percentage increase in deterrence.

Robustness of the patience-free regime. When $\beta_\alpha \geq \beta_D$, $\delta^*(q) = 0$ for every $q < 1$: if the mediator's coordination value exceeds the deviation temptation, cooperation is individually rational in the stage game and the action margin requires neither patience nor monitoring. This robustness is why the policy discussion below centers on reaching the regime under monitoring uncertainty (Section 9).

Maximum tolerable detection failure. For $g > 0$ and fixed $\delta \in (0, 1)$, inverting (5) gives

$$\bar{q}(\delta) = 1 - \frac{(1-\delta)g}{\delta\kappa_{\text{eff}}}, \quad (9)$$

and the admissible set of detection failures is $\mathcal{Q}(\delta) = [0, \bar{q}(\delta)]$ when $\bar{q}(\delta) \geq 0$ and $\mathcal{Q}(\delta) = \emptyset$ when $\bar{q}(\delta) < 0$: a negative $\bar{q}$ means δ lies below even the perfect-monitoring threshold $\delta^*(0)$, so no monitoring quality rescues cooperation, and clipping it to zero would falsely suggest that perfect monitoring suffices. For $g \leq 0$ every $q < 1$ is admissible. Where nonempty, $\bar{q}(\delta)$ is a measurable upper bound on how unreliable external-action monitoring may be before coordination collapses at a given patience level, directly testable in deployed systems.

Finally, for $g > 0$ the two design levers are not symmetric *per unit of the normalized payoff coordinates* (in the patience-free region both derivatives vanish): since β_α enters $\delta^*(q)$ twice (narrowing the gap g and enlarging the punishment swing κ_{eff}), a unit improvement in β_α relaxes the patience requirement more than a unit increase in β_K . Whether this ordering survives in engineering units depends on the calibration maps and implementation costs of the two levers, which the model does not fix:

$$\frac{|\partial\delta^*/\partial\beta_\alpha|}{|\partial\delta^*/\partial\beta_K|} = \frac{g + \kappa_{\text{eff}}}{g} > 1, \quad \text{independent of } q. \quad (10)$$

The two levers are substitutes, not complements, in relaxing patience requirements (Appendix A.4). This dominance concerns the patience threshold specifically; Section 8 shows that once the patience conditions are comfortably satisfied, the *participation* margin governs most of the remaining uncertainty about stability.

6.3 C2*: ex-ante participation and commitment

Stability also requires that each agent prefer membership in the oversight coalition to removing it. Assume transferable utility (side payments are feasible). Under the additive oversight value $v(S) = |S| \beta_\Omega$, symmetry gives every agent the equal split $\phi_i = \beta_\Omega$ per period, which is simultaneously the Shapley value (Shapley 1953) and the unique core allocation (Bondareva 1963); on the symmetric path the transfers are identically zero, and the cooperative-game apparatus matters only off that baseline. A member who stays on the cooperative path receives $\beta_\alpha + \beta_\Omega$ every period; removing the mechanism yields the one-time gain β_ℓ/N , forfeits both flows, and returns play to the zero-normalized baseline of Section 3.1. The exact participation condition, necessary and sufficient for this membership decision under zero outside continuation value, is therefore $(\beta_\alpha + \beta_\Omega)/(1 - \delta) \geq \beta_\ell/N$. The $1/N$ share follows from the baseline assumptions of Section 3.1: removal releases a rival fixed pool that is divided equally, so a lone remover captures only its own share of the total gain β_ℓ . Non-excludability alone does not imply this dilution; Appendix C treats the non-rival and bounty alternatives. As our reported condition we deliberately drop the forfeited coordination flow and require the membership flow alone to carry the comparison:

$$C2^* : \quad \frac{\beta_\Omega}{1 - \delta} \geq \frac{\beta_\ell}{N} \iff \beta_\Omega \geq (1 - \delta) \frac{\beta_\ell}{N}.$$

C2* is sufficient but not necessary: whenever it holds the exact condition holds a fortiori, since $\beta_\alpha \geq 0$, so every stability region computed from C2* is a lower bound on the exact region. The relationship is sharper than generic conservatism. Let the outside flow after removal be $v_{\text{out}} \in [0, \beta_\alpha]$, measuring how much of the coordination value survives the mechanism's loss; participation then requires $(\beta_\alpha + \beta_\Omega - v_{\text{out}})/(1 - \delta) \geq \beta_\ell/N$. The exact zero-baseline condition is the case $v_{\text{out}} = 0$, and C2* is exactly the case $v_{\text{out}} = \beta_\alpha$: C2* certifies participation against every outside option that preserves at most the coordination flow, and we accordingly call the volumes computed from it *certified*. The distinction matters quantitatively. Under the declared ranges the zero-baseline condition never binds ($\min(\beta_\alpha + \beta_\Omega) = 0.35$ exceeds $\max((1 - \delta)\beta_\ell/N) = 0.225$), so exact-model stability reduces to C1** alone (97.9%, independent of N), and every participation and group-size result in this paper is a statement about robustness to the outside option, not about fragility of the zero-baseline model. What removal actually destroys is therefore a structural uncertainty on par with the stipulated ranges, and it joins the epistemic ledger. Setting $\delta = 0$ recovers the strictest form $\beta_\Omega \geq \beta_\ell/N$, the no-future-value comparison in which a single period's membership flow must outweigh the removal share on its own; we carry the present-value form through all computations and quote the $\delta = 0$ bound only where a δ -free statement is convenient. The dropped coordination flow is typically 1.5 to 10 times β_Ω under the declared ranges (5th–95th percentile of $\beta_\alpha/\beta_\Omega$; the envelope is 1 to 18), so the conservatism is substantial: C2* materially overstates the attraction of removal, and the reported volumes correspondingly understate participation. As N grows, the per-agent removal gain shrinks, a dilution effect that eases participation; the effect is a property of the rival fixed-pool reading of β_ℓ , and Appendix C shows it vanishes for non-rival removal benefits and can reverse under remover-exclusive bounties; coalition removal structures are covered by A_{dev} and Appendix C.

6.4 Joint sufficiency, and what is necessary

Jointly the conditions are sufficient: given C1**(q), no agent gains from any single-period deviation from the cooperative path (and hence, by the one-shot deviation principle for public strategies, from any deviation plan). Given C2*, membership has higher present value than removal for every agent, and by A_{dev} equal-share coalitions face the same per-member comparison, so no baseline coalition improves on it (appropriation and non-rival structures are outside the theorem; Appendix C). Under immediate enforcement, C1*(q) additionally makes following the recommendation a stage-game best response, so cooperation survives even for agents that discount the future entirely. Necessity is asymmetric. C1**(q) is necessary: without it the deviation value V_D exceeds V_C and the grim-trigger profile is not an equilibrium. For voluntary membership, necessity attaches to the exact

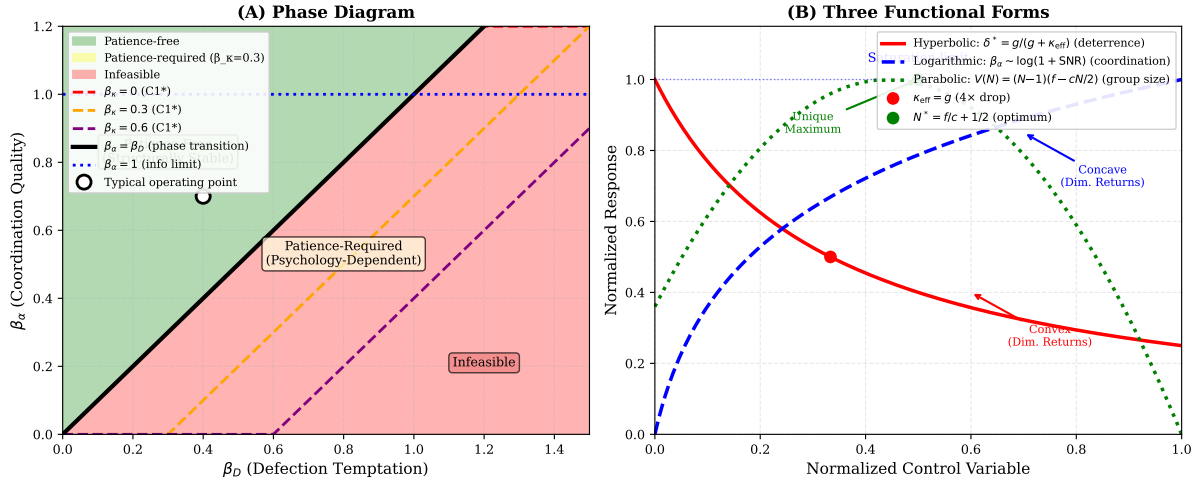


Fig. 1. **Functional structure of the stability conditions.** (a) Phase diagram: patience-free regime ($\beta_\alpha \geq \beta_D$), patience-required regime, and infeasible region. (b) Three functional forms: hyperbolic deterrence ($\delta^* = g/(g + \kappa_{\text{eff}})$ at $q = 0$), logarithmic coordination saturation ($\beta_\alpha \sim \log(1 + \text{SNR})$), and parabolic group value ($V(N) = (N-1)(f - cN/2)$). Derivations in Appendix A.

condition $(\beta_\alpha + \beta_\Omega)/(1 - \delta) \geq \beta_t/N$; the reported C2* is its conservative sufficient strengthening (Section 6.3). C1*(q) is not necessary. For example, $\beta_\alpha = 0.3$, $\beta_D = 0.9$, $\beta_k = 0.5$, $q = 0.3$, $\delta = 0.95$ (all within Table 1) violates C1*(q) ($0.65 < 0.9$), yet $\delta^*(q) = 0.517 < \delta$ and deviation does not pay: the one-shot deviation value $\beta_D + \delta[(1 - q)V_P + qV_C] = -4.04$ and the perpetual-deviation value $V_D = -8.04$ both fall far below $V_C = 6.0$, so the continuation threat alone carries cooperation. What C1*(q) buys is independence from δ ; where it holds, cooperation is robust to any discounting, so the phase diagram (Fig. 1a) separates the structurally stable region from the psychology-dependent one. The all-defect profile is an equilibrium of the underlying game under the stage-Nash assumption of Section 3.1; the mechanism's role is coordination of public continuation strategies, not enforcement. No necessity claim is made across all public equilibria, punishment schemes, or removal technologies. $\square$

What the conditions buy back in the hedging game. The coordination game and the hedging game of Section 4 connect through the interference technology. A mechanism satisfying the three conditions makes detected exploitation trigger coordinated retaliation, which in the notation of Section 4 raises the survival probability σ of an undefended agent and lowers the loss L when interference does occur. Both changes raise the non-hedger payoff at every p , so the equilibrium hedging level falls. It does not reach zero: Lemma 4.1 continues to apply because assumptions (i)–(iii) are properties of the physical environment, not of the mechanism. The way out that the opening paragraph promised is therefore a way out of what the arms race costs, not out of hedging itself: coordination shrinks the defensive diversion that physics makes unavoidable. How many agents that coordination should span is the question the theorem leaves open.

7 Optimal Group Size Depends on Benefit Structure

Theorem 6.1 leaves group size N as a free parameter, and two effects compete over it. Adding agents improves stability through exactly one channel in this model: the per-member removal gain β_t/N shrinks under the rival fixed-pool assumption (q and β_k are sampled independently of N , so no monitoring-coverage effect is modeled).

Imperfect Monitoring: Patience Requirement and Stability Mass

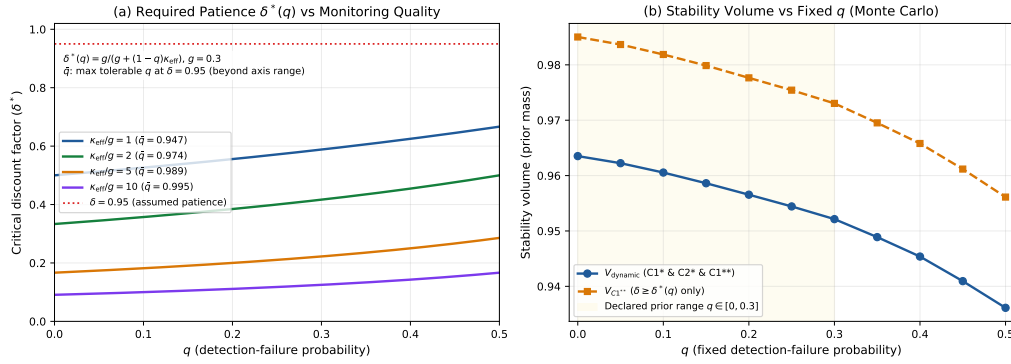


Fig. 2. **Monitoring quality and sustainability.** (a) Critical discount factor $\delta^*(q)$ as detection failure q grows, for several ratios κ_{eff}/g ; the maximum tolerable detection failure $\tilde{q}$ of (9) at the feasibility bound $\delta = 0.95$ is annotated per curve (all $\tilde{q} > 0.94$, beyond the plotted range). (b) Monte Carlo stability volume as a function of fixed q under baseline parameter ranges.

Against this, coordination costs scale quadratically: all-to-all communication requires $N(N-1)/2$ channels, and consensus overhead grows at least linearly (Castro and Liskov 1999; Lamport 2001). The quadratic charge prices the trust-minimized implementation, in which the agents maintain the common-state layer among themselves by Byzantine agreement rather than trusting any single operator (the guarding-the-guardian concern of Section 9); with a single trusted mediator, communication is $O(N)$ and the cost model below, including its optimum N^* , does not apply, while subquadratic agreement protocols would soften the tradeoff, as falsifiable prediction (3) anticipates (Section 10). To quantify this tradeoff, we define the *certified* stability volume as the parameter-space mass of the theorem’s own equilibrium object, the continuation-enforcement pair with the conservative participation bound, under uniform priors:

$$V_{\text{dynamic}} = \Pr[C1^{**}(q) \wedge C2^*], \quad (11)$$

with all eight parameters, including q , drawn from the ranges of Table 1. The immediate-enforcement regime is summarized by two quantities, reported alongside where relevant: its full equilibrium volume $V_{\text{immediate}} = \Pr[\delta \geq \delta^*(q, s) \wedge C2^*] = 97.8\%$ (the threshold of Section 6.1 at $\rho = 1$), and the myopic benchmark $V_{\text{myopic}} = \Pr[C1^*(q) \wedge C2^*] = 97.5\%$, the mass on which action obedience needs no patience at all. Each volume describes one mechanism, and we do not intersect the regimes (the conservative three-way intersection, 95.6%, appears in the reproduction package for reference only). Both are *certified* volumes in the sense of Section 6.3. With the zero-baseline participation condition instead, which never binds in the declared ranges, exact-model stability is $\Pr[C1^{**}] = 97.9\%$ for every N , and the full immediate regime has $\Pr[\delta \geq \delta^*(q, s)] = 99.99\%$, with $\Pr[C1^*] = 99.7\%$ its myopic subset. All N -dependence below therefore lives on the certified margin. These volumes are conditional illustrations, not measurements (Section 10).

Monte Carlo evaluation (Appendix E) shows certified stability rising monotonically with group size (Fig. 3). At $N=2$ to 3, the certified participation margin binds hardest and V_{dynamic} ranges from 86.3% to 93.4%: small coalitions are the fragile case on that margin because each member’s share of the removal gain β_t/N is largest there. The 95% level is crossed at $N=4$ (96.1%), and V_{dynamic} saturates near 97.9% at $N=10$, with marginal gains

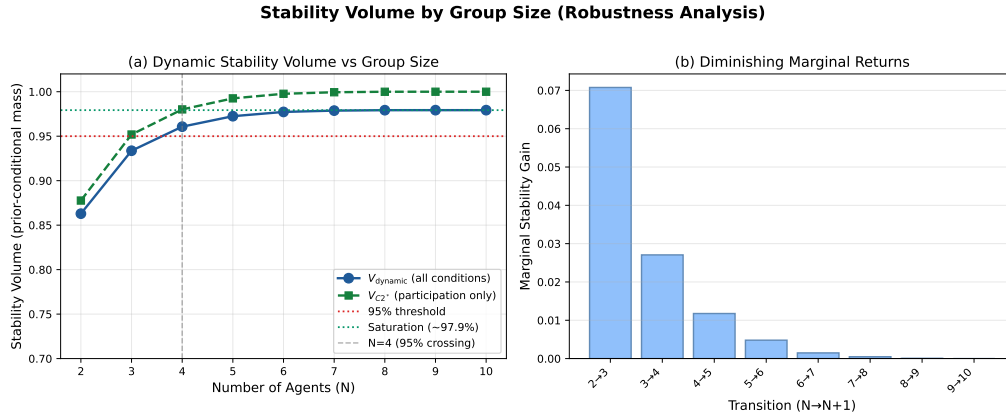


Fig. 3. **Stability volume versus group size.** (a) V_{dynamic} (solid) and participation constraint V_{C2^*} (dashed) versus N . (b) Marginal gains shrink with N and fall below one percentage point beyond $N=5$. Monte Carlo details in Appendix E.

below one percentage point per additional agent beyond $N=5$. Stability volume alone would still favor larger groups, but coordination costs pull the other way, and how the two reconcile depends on how benefits scale.

Total coalition value, with each member beyond the first adding benefit f and each pair costing c , is $V(N) = (N-1)f - c\binom{N}{2} = (N-1)(f - cN/2)$, yielding the continuous optimum $N^* = f/c + \frac{1}{2}$, with the discrete optimum at one of the two neighboring integers (derivation in Appendix A). The result is sensitive to how benefits scale (Fig. 4): saturating benefits (e.g., redundant verification with diminishing returns) give $N^*=2$; linear benefits (e.g., independent monitoring domains) give $N^*=3$ to 5; and network effects (e.g., shared threat intelligence with cross-validation gains) give $N^*=8$ to 10. Optimal group size is therefore a design variable contingent on benefit structure, not a universal constant. Under linear benefits, the cost-optimal range $N^*=3$ to 5 buys 93.4% to 97.3% stability volume, so the cost-optimal sizes and the $\geq 95\%$ corridor overlap at $N=4$ to 5: the group sizes that are cheapest to coordinate are also stable ones. Only the smallest coalitions fall short, and there participation incentives (raising β_Ω) are the direct remedy (Section 8).

8 What Drives Certified Stability: Patience and Participation Lead, Monitoring Imperfection Is Cheap

The stability conditions involve eight parameters of unequal epistemic standing, and this section asks which of them the conclusions actually depend on. Some carry hard bounds: $\hat{\alpha}_M \leq 1$ by construction (so $\beta_\alpha \leq 1$ under the identity calibration of Section 3.3), $\delta \leq 1$ and $q \in [0, 1]$ by definition. Others are engineering estimates with significant uncertainty. Under the baseline ranges of Table 1, the certified continuation volume is $V_{\text{dynamic}} = 95.8\%$, with per-condition volumes $V_{C1^{**}} = 97.9\%$ and $V_{C2^*} = 97.8\%$; the immediate-enforcement regime has $V_{C1^*} = 99.7\%$, full equilibrium volume $V_{\text{immediate}} = 97.8\%$, and myopic benchmark $V_{\text{myopic}} = 97.5\%$. The pattern is informative: the patience and participation conditions are satisfied over comparably large shares of the space, with participation binding hardest at small group sizes. One number deserves emphasis: 57.9% of the declared space satisfies $\beta_\alpha \geq \beta_D$ outright, sitting in the patience-free regime where the action margin needs neither the discount factor nor monitoring quality.

The cost of imperfect monitoring is real but modest. Sampling q over its engineering range $[0, 0.3]$ costs half a percentage point of stability volume relative to the perfect-monitoring baseline (95.8% versus 96.3%); pushing q

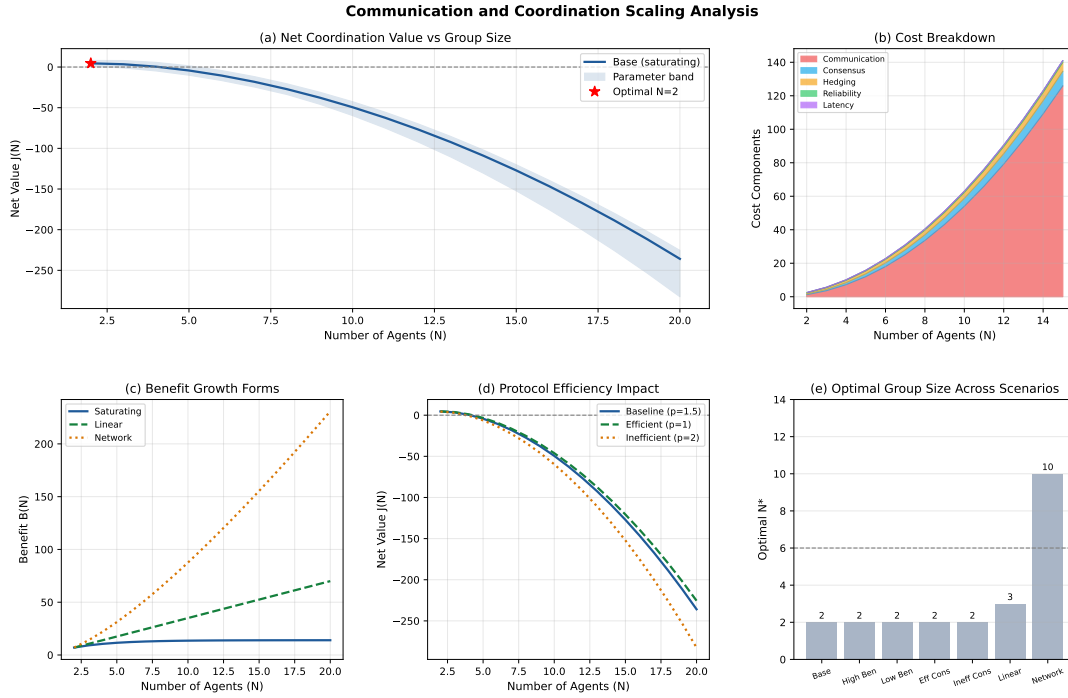


Fig. 4. Benefit structure determines optimal group size. (a) Net value $J(N)$ for saturating benefits. (b) Cost breakdown. (c) Benefit forms compared. (d) Consensus efficiency variants. (e) N^* at representative parameters per scenario: saturating ($N^*=2$), linear ($N^*=3$), network ($N^*=10$); across the parameter ranges these scenarios span $N^*=3$ to 5 (linear) and 8 to 10 (network). Derivations in Appendix A.

to a fixed 0.3 or 0.5 lowers V_{dynamic} to 95.1% and 93.5%, respectively (Fig. 2b). Monitoring imperfection therefore matters as the closed form predicts: continuously, through the discounted deterrence $(1 - q)\kappa_{\text{eff}}$, with collapse only as $q \rightarrow 1$, but within the realistic range for external-action monitoring it does not destabilize the mechanism. The reason is structural: most of the parameter space sits either in the patience-free regime (where q is irrelevant by the robustness result of Section 6.2) or far from the $\delta^*(q)$ boundary. Perfect monitoring is in any case not the deep requirement. What matters is that the public signal distinguish deviation from compliance: for a binary monitoring technology with false-negative rate q and false-alarm rate r , informativeness of the deviation-punishing rule requires $q + r < 1$, and as $q + r$ approaches one, detected events cease to provide useful evidence for selecting continuation states (beyond that boundary the signal is anti-informative and relabeling restores it; Appendix C.2). The main text sets $r = 0$; the false-alarm extension of Appendix C.2 demonstrates the boundary.

Patience and participation lead the uncertainty budget. Varying each parameter's range by $\pm 20\%$ and recomputing V_{dynamic} (full table in Appendix E) identifies the discount factor δ as the strongest single lever on the certified stability volume, with sensitivity coefficient 0.13, followed by the oversight value β_{Ω} (0.10); group size and the stage-game payoffs cluster near 0.05, coordination quality within its baseline range contributes little (0.02), and detection failure least of all (0.004). Sobol variance decomposition over all eight parameters (Appendix E) answers the complementary question of what drives the certified stability *margin* (the exact condition

does not bind; Section 6.3): β_Ω leads total-order indices at every group size, with δ second throughout (Fig. 5). The two rankings are consistent, not contradictory: δ moves the probability of stability most because it enters both the patience threshold and the present-value participation constraint, while β_Ω sets the level of the binding margin itself. That the least-grounded parameters top both rankings is not an embarrassment the analysis hides but the finding it delivers: the framework does not claim β_Ω and δ are known; it identifies them as what must be measured before any deployment leans on these results. Two qualifications attach to these rankings. Both are conditional on the declared ranges and independence assumptions and need not persist under wider monitoring-error ranges, correlated priors, alternative removal-benefit structures, or explicit false positives. And they are global variance-importance statements, to be distinguished from the local analytic comparisons of Section 6.2 (such as the lever ratio $(g + \kappa_{\text{eff}})/g$), which are derivative results holding at every parameter point regardless of ranges.

The closed form and the global analysis disagree about what matters, and the disagreement is informative: the lever the theory spotlights, coordination quality, is neither the strongest lever on the stability volume nor the main carrier of margin variance. Two lessons follow, one empirical and one about mechanism design. Empirically, the highest-priority target is the outside option itself, since whether removal destroys the coordination it enabled decides whether the participation margin exists at all (Section 6.3). After it come the oversight value that agents actually derive from membership (β_Ω) and the effective discount factor δ , which for an AI system is not directly observable and must be inferred from behavior. The coordination quality β_α and the detection-failure probability q of deployed monitoring channels complete the list at the patience boundary. For design, the two analyses assign the levers different jobs: β_α is the strongest lever for *eliminating patience and monitoring requirements* (entering the patience-free regime), while β_Ω is the efficient lever for *keeping coalitions together* once patience requirements are met. Question 2 thus has its answer: a conditional yes, on the terms just quantified. With the stability conditions, their parameter sensitivity, and optimal group size characterized, one institutional issue remains: what kind of entity can mediate among systems that exceed any human institution in computational capability?

9 The Weak Leviathan: Capability-Independent Public-State Mediation

The classical answer to who enforces cooperation, from Hobbes's *Leviathan* (Hobbes 1651) to modern enforcement theory, requires a central authority more powerful than the agents it governs. No human institution satisfies this requirement for advanced AI systems: human cognition operates at $\sim$ kHz, six orders of magnitude below electronic circuits at $\sim$ GHz; the public actions that matter for monitoring arrive at $\sim$ 1 per second, slower still. Mediation appears necessary but the mediator inadequate; this is the impasse Question 3 names. The framework resolves it by rejecting the premise: mediation requires no superior capability, and observable weakness is what makes a mediator affordable to tolerate and cheap to inspect; credibility itself comes from commitment, authentication, and non-equivocation (Proposition 9.1), which weakness makes easier to verify, not unnecessary. The institutions whose weakening the opening paragraph recorded are not, on this analysis, disqualified from the mediating role; the weakness that matters, the inability to enforce, is what the role requires.

PROPOSITION 9.1 (CAPABILITY-INDEPENDENT PUBLIC-STATE MEDIATION). *Let the institutional architecture consist of three separated layers: the mediator M , a committed, non-strategic automaton implementing the public state machine $z_{t+1} = T(z_t, y_t)$ of Section 6; the commitment layer C of Section 3.2, which makes stage-0 membership irreversible (A_{commit}); and, in the immediate-enforcement regime, a separate automated sanction layer E (escrow, access control) whose triggers are deterministic functions of the authenticated public signal y_t , which under A_{ext} carries the deviator's identity; the binary state z_t alone could not identify whom to sanction. Then the architecture (C, M, E) supports the equilibrium of Theorem 6.1 or Corollary 6.2 among agents $\{A_1, \dots, A_N\}$, with M supplying the authenticated public-state layer, C supplying irreversible commitment, and E executing available sanctions, if:*

Summary: Coordination Stability Analysis

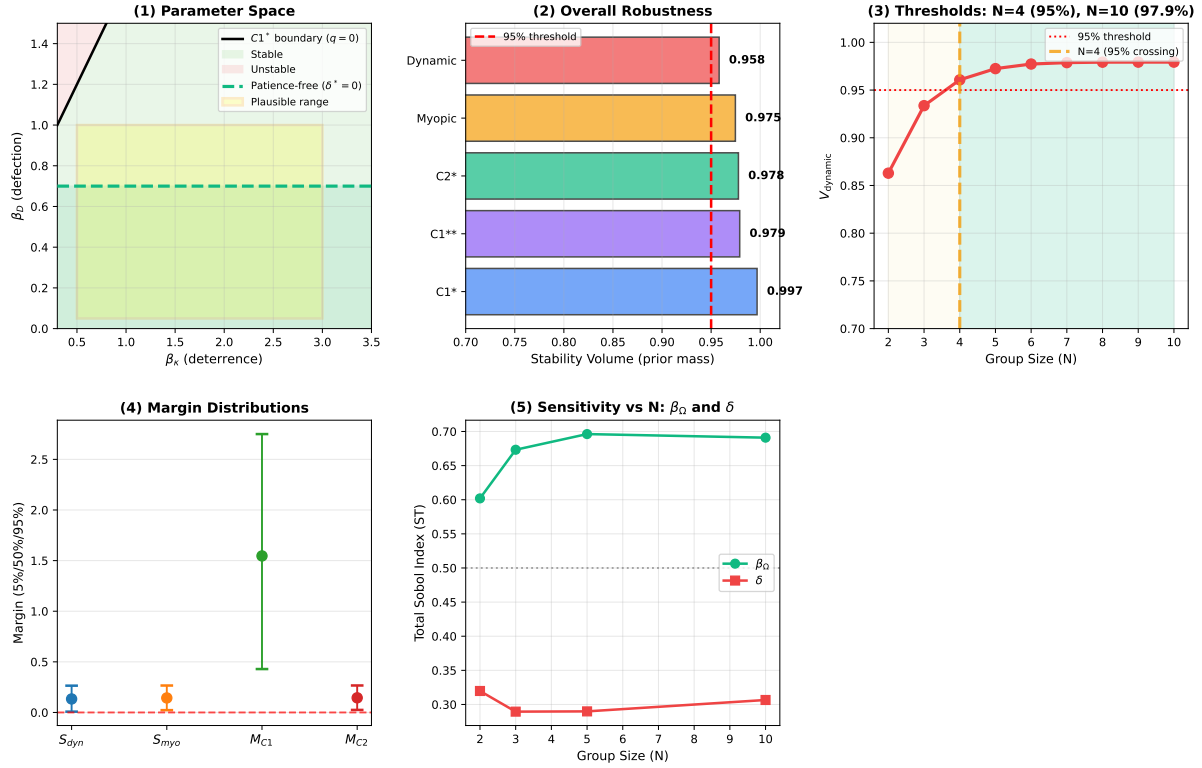


Fig. 5. **Analysis overview.** (1) $C1^*$ stability boundary in the (β_k, β_D) plane at $q = 0$. (2) Stability volumes (one-shot to dynamic). (3) $V_{dynamic}$ versus N . (4) Stability margin distributions. (5) Sobol indices: β_O leads at every N , with δ second throughout. Extended Sobol analysis in Appendix E.

(i) common state: all compliant agents derive the same authenticated state z_t , e.g., from an append-only public log or a reliable-broadcast protocol, so that z_t is common knowledge among them (Halpern and Moses 1990); (ii) informative action monitoring: the agent-deviation channel separately satisfies A_{ext} (this channel is distinct from state distribution; a low per-agent message-failure rate would not by itself yield common knowledge (Halpern and Moses 1990)); (iii) weak capacity: M has no enforcement or commitment capability of its own; those reside in E and C ; and (iv) stability: under continuation enforcement, $C1^{**}(q)$ and $C2^*$ hold; under immediate enforcement via E , $\delta \geq \delta^*(q, s)$ and $C2^*$ hold (Corollary 6.2; $C1^*(q)$, equivalently $g_s = 0$, identifies the myopic subset).

PROOF. By (i), the public state is common knowledge among compliant agents; by (ii), deviations enter the public state with detection failure q . Theorem 6.1 then applies with M 's log as the public state: membership irreversibility is supplied by C (A_{commit}), so the stage-0 join decision priced by $C2^*$ is the only participation decision, and by (iv) the grim-trigger profile with equal-split participation transfers is an equilibrium of the game among the agents. M introduces no strategic moves because it is not a player: its commitment to T is an assumption about the institution, not a consequence of its weakness, and condition (iii) matters for a different reason, namely that an M without enforcement capability is one the agents can afford to tolerate and can cheaply

verify (Section 5.1). In the immediate-enforcement regime the payoff-changing sanction is executed by E as a deterministic function of the authenticated public signal (which carries the deviator's identity under A_{ext}), so no discretionary power accrues to M and (iii) is preserved. The mediator does not verify intentions, which cannot rest on exact real-time simulation between saturated peers (Theorem 5.1) and which the specification and binding gaps impede more broadly; it maintains common knowledge of the state and, through the agents' own monitoring, of compliance. $\square$

Stated plainly, the mechanism is modest: on the equilibrium path the institution is a bulletin board with a reputation, the deterrence is done entirely by the agents, and it governs as a notary governs. Three caveats bound the modest reading. First, the weakness claim is about M alone: the layers C and E are strong in the narrow, mechanical sense of being irrevocable and automatic, and the mediator's weakness must not be mistaken for weakness of the institution entire. Second, the committed-automaton assumption does real work: equilibrium selection is itself a form of power, and an operator that deviated from T (steering among equilibria, delaying announcements, logging entries strategically) would violate the assumption rather than the theorem; an append-only log makes such deviations evident after the fact but does not prevent them, so guarding the guardian reappears one level up as an institutional-design problem (log transparency, redundancy across institutions), open rather than solved. Third, the claim is conditional throughout: human institutions need not dominate advanced systems to mediate among them *provided* the agent-side conditions of Theorem 6.1 already hold; the institution supplies selection, not stability.

The observability asymmetry from Section 5.1 makes condition (i) cheap to implement rather than supplying it outright: when the institution is human, its state changes arrive at ~ 1 per second (the machine $\rightarrow$ human row of Table 3), so machine agents can read, cross-check, and archive an authenticated institutional log at negligible cost, which is what an append-only common-state layer requires. Human limitations become advantages: slow processing is non-threatening (condition iii), physical vulnerability creates aligned incentives against catastrophic conflict, and inability to rapidly self-improve makes institutional commitments easier to inspect and harder to silently rewrite, though commitment itself remains an assumption (Proposition 9.1).

Human institutions thus serve two distinct roles, neither requiring superior capability. First, observable human actions function as Schelling focal points (Schelling 1960): even computationally inferior signals create common knowledge when they are publicly observable, providing the coordination function that the public state machine requires. Second, the capacity to impose civilization-scale costs, even if mutually harmful, raises the effective conflict cost β_k , reducing patience requirements via the hyperbolic relationship (5). The two roles do not conflict with condition (iii) of Proposition 9.1: (iii) constrains the *mediator function*, which must be unable to enforce outcomes so that its signals change no payoffs, while deterrence operates through the environment the agents inhabit, of which human infrastructure control is one part; the mediating institution need not be, and should not be, the entity wielding it. This mechanism has a temporal structure: during an initial phase of infrastructure dependence, human-controlled compute, energy, and communication substrates provide conventional policy levers that can shape parameters into the stable region. Once autonomous operation begins, these levers weaken, making the early-phase parameter configuration a non-renewable resource. The window claim is a motivated premise about the environment, not a result derived within the model; population-dynamics models of the AI race reach the analogous conclusion that regulation helps only when timed to the race's dynamics (Han et al. 2020).

9.1 Policy design principles: preserving human control

The mechanism condenses into a doctrine, stated once: human control does not require cognitive superiority. It requires that consequential AI actions remain attributable, that participation in shared infrastructure remains valuable, and that access can be withdrawn faster than a deviation becomes irreversible (in the model: within

the period). Four design principles follow, each tied to a quantity the model prices; empirical effort to measure those quantities should follow the ordering of Section 8.

Preserve infrastructure dependence. The commitment layer C and the sanction layer E exist only while systems act through resources that accountable institutions control: compute, energy, network access, credentials, settlement. During that phase the conventional levers work: proliferation controls bound N , architectural requirements raise β_α and lower β_D , infrastructure dependencies raise β_Ω , and monitoring standards cap q . A system that becomes simultaneously autonomous in compute, energy, communication, and replication removes the substrate on which C and E operate; by the window premise above, governance designed after that point has fewer levers, so the ordering is itself the principle: govern before autonomous operation, not after.

Regulate observable actions, not unverifiable intentions. A_{int} says internal states resist verification; A_{ext} says external actions do not. Policy conditioned on claims about intent inherits the first problem; policy conditioned on authenticated actions (deployment, resource acquisition, replication, high-impact execution) inherits the second's tractability. Detection infrastructure is then a quantifiable substitute for deterrence (Section 6.2), and the tolerable-failure bound $\bar{q}(\delta)$ gives monitoring procurement a concrete reliability target.

Make compliance the profitable route. The most robust regime is patience-free: raise coordination value above the deviation temptation ($\beta_\alpha \geq \beta_D$), which by Section 6.2 also removes the dependence on monitoring quality. Signal quality is the binding constraint, since learning agents reach efficient correlated equilibria from even simple shared signals (Cigler and Faltings 2013); the membership flow β_Ω is raised by the services the institution bundles (shared infrastructure, trusted counterparties, data, standards, market access). The same design must keep removal unattractive: the participation condition of Section 6.3 fails exactly when destroying the mechanism beats membership, so replicated logs, redundancy across jurisdictions, and no single point of unilateral disabling protect the margin. Optimal group size remains benefit-structure-dependent (Section 7): characterize whether coordination yields saturating, linear, or network returns before selecting N .

Separate mediation from enforcement, and precommit consequences at machine speed. The architecture is (C, M, E) , not an omnipotent regulator: the mediator stays non-enforcing and cheap to inspect, while sanctions execute automatically from the authenticated record (the immediate regime of Corollary 6.2). Human legitimacy sets and audits the rules; machine-speed mechanisms execute them within the period, which is what withdrawal “faster than a deviation becomes irreversible” requires. Slow, deliberative punishment instead returns the system to the continuation regime and its patience requirement, restoring the dependence that machine-native enforcement removes.

10 Discussion

The introduction posed three questions. The first receives a harder answer than optimistic intuition suggests: zero hedging is evolutionarily unstable under interference conditions that physical constraints make generic (Lemma 4.1), and full real-time peer simulation exceeds any equal monitor's budget (Theorem 5.1). Capability does not substitute for trust. The second receives a conditional yes: sustainability (necessary and sufficient for the specified grim-trigger public strategy against unilateral deviations) together with a conservative participation bound suffices for coordination built on observable actions alone, with one-period deterrence the strengthening that frees action obedience from patience (Theorem 6.1). All of this holds under realistic, imperfect monitoring; certified participation incentives are the binding margin at small group sizes, and group size is a benefit-structure-dependent design variable. The third receives a more permissive answer than the classical tradition expects: mediation requires observability and consistency rather than superior capability, so observable weakness is a viable basis for mediation (Proposition 9.1). Nothing in the stability conditions is specific to artificial agents: they apply to any strategic actors whose internal states are opaque and whose external actions are observable; their structure echoes arms control and cartel stability for that reason.

What the percentages mean. The stability volumes reported above are conditional illustrations, not measurements. They answer one question: over the declared parameter ranges, how much of the space satisfies the conditions of the regime being reported (V_{dynamic} for continuation enforcement, $V_{\text{immediate}}$ for immediate enforcement)? Change the ranges, or the modeling of a constraint, and the digits change with them. This caution is earned rather than rhetorical: correcting the participation constraint to its present-value form during revision moved the headline volume by several points and reordered the sensitivity ranking. What survived that correction, and every rerun and range variation, is the structure: the primacy of the patience-free regime, and monitoring imperfection within its realistic range being nearly free. The group-size structure is a certified-margin property: stability rises with N on the worst-case margin, while under the zero-baseline exact condition participation never binds and stability is N -independent (Section 6.3). A reader should carry away the structure, its conditions, and not the percentages.

Limitations. Five limitations qualify these results. First, parameter ranges mix hard physical bounds (e.g., $\hat{\alpha}_{\text{MI}} \leq 1$ from Shannon theory, giving $\beta_{\alpha} \leq 1$ only under the identity calibration) with engineering estimates carrying significant uncertainty; the participation parameters β_{Ω} and β_{ℓ} are stipulated outright (Appendix A) and sit on one of the margins the sensitivity analysis ranks most consequential, which makes them the paper's main grounding gap. The range-sensitivity analysis quantifies, but does not eliminate, this dependence. Second, V_{dynamic} assumes uniform priors over parameter ranges; alternative priors yield different stability volumes. Third, the monitoring model is the simplest that captures detection failure: misses are i.i.d. public events and false alarms are set to zero in the main text. With false alarms, grim trigger destroys surplus on the equilibrium path, and finite punishments in the style of Green and Porter (1984) can be preferable when incentive-compatible; Appendix C.2 derives the corresponding sustainability condition and shows our threshold remains the $\varepsilon \rightarrow 0$ benchmark. Private or correlated monitoring structures (Kandori and Obara 2006; Sugaya and Wolitzky 2017) are beyond our scope, and real deployments will add failure modes the clean model excludes: correlated blind spots, attribution disputes, adversarial attacks on the monitoring infrastructure itself, and mediator capture (Section 9). Hidden capability building and undetectable goal drift, which no external-action observability can address, remain the framework's boundary conditions (Appendix F). Fourth, grim trigger is not renegotiation-proof; we use it because it gives the weakest patience requirement among the punishment automata considered here; renegotiation-proofness generally weakens punishments and may tighten the threshold, and we do not characterize that refinement. Fifth, the framework addresses coordination stability given alignment uncertainty, not alignment itself: it does not claim that coordination solves the alignment problem, but complements alignment research with redundancy and monitoring; and all quantitative predictions are conditional on the multi-agent scenario arising at all. These limitations are also the work plan: richer monitoring structures, renegotiation-proof punishments, explicit coalition formation, and alternative priors are the natural next steps, and each attaches to a specific assumption declared above.

Falsifiable predictions. The framework yields four falsifiable predictions. (1) Because C2* is sufficient rather than necessary, stable small coalitions do not by themselves falsify the model; the testable version is conditional: if the outside option is independently estimated at $v_{\text{out}} \approx \beta_{\alpha}$ and voluntary coalitions with $N < 4$ nevertheless sustain stability while violating C2*, the modeled removal-payoff specification is rejected. (2) If AI systems develop peer verification of internal states with detection failure $q < 0.1$, then the simulation-incompleteness assumption is circumventable in practice and A_{int} fails. (3) If linear-benefit institutions with $N = 15$ to 20 sustain flat coordination for more than 30 years, then the quadratic-cost assumption fails, likely indicating subquadratic communication protocols. (4) If systems satisfying $\beta_{\alpha} > \beta_D$ systematically fail to cooperate in one-shot settings, then C1*(q) is insufficient and additional mechanisms (reputation, trust, institutional commitment) are required. Detailed operationalizations appear in Appendix F. The four predictions are not equally strong tests: (1) and (3) require multi-decade observation windows, and (2) is a bet on future verification technology. Prediction (4) is

testable now, and cheaply, so we tested it (Appendix G): in 480 one-shot games among two current language-model agents with β_α and β_D engineered into the payoffs, cooperation sits at 91–94% in the patience-free regime ($\beta_\alpha > \beta_D$, with or without a mediator recommendation) and falls to 13–40% beyond the boundary. Prediction 4 is not falsified, and the boundary the theory identifies is where behavior actually breaks. One result exceeds what the one-shot theory licenses and we report it as a caution rather than a confirmation: in the dilemma region, where the recommendation is not incentive-compatible, the non-enforcing mediator still tripled cooperation (12.8% to 40.4%), so current language-model agents deviate from the rational-agent model in the direction of excess compliance. The effective discount factor can likewise be elicited behaviorally from switch points between immediate and delayed rewards (Table 5); whether the elicited δ is even stationary is itself informative.

Practical utility. For designers of multi-agent AI deployments, the conditions translate into measurable engineering targets: the coordination quality β_α (target: above the deviation temptation, to enter the patience-free regime); the detection-failure probability q of action monitoring (target: below $\bar{q}(\delta)$ of (9)); and per-member oversight value above $(1 - \delta) \beta_\ell / N$ for the certified condition, with the general outside-option target $\beta_\alpha + \beta_\Omega - v_{\text{out}} \geq (1 - \delta) \beta_\ell / N$. For policymakers, the analysis identifies which levers matter during the window of infrastructure dependence; within the normalized payoff coordinates and for $g > 0$, signal quality dominates deterrence on the patience margin (10), but a full ranking across signal quality, monitoring, and deterrence would require a common cost normalization the model does not fix. A stylized calibration showing every parameter measurable from existing telemetry is given in Appendix F (Table 5).

11 Conclusion

Stable coordination among autonomous AI systems is possible in principle. It is constrained by physical law and game-theoretic structure, not by optimistic assumptions about alignment verification, and it is far from automatic. Five structural insights survive every parameter variation examined even where specific percentages do not: some defensive hedging is always selected, so coordination shrinks the arms-race diversion rather than abolishing it; simulation incompleteness motivates external mediation; imperfect monitoring discounts deterrence smoothly, by exactly the detection probability, and never destabilizes the patience-free regime; group size is contingent on benefit structure; and a weak institution can serve as an authenticated public state machine, so that mediation rests on observable weakness rather than on power, conditional on the agent-side incentives already holding. One asymmetry runs through all five: internal computation is opaque even to an equal, while external action is visible even to the weak. It is why coordination must be built on actions rather than minds, and why the institution that keeps the common record of those actions can be weak. Whether the possibility becomes reality depends on the current window of infrastructure dependence, in which policy and design can still shape coordination parameters, monitoring reliability among them, into the stable region, along the principles of Section 9.1. Once autonomous operation begins, that window closes, and it does not reopen. The experiment may run only once; the point of the analysis is to know the way out before it runs.

Acknowledgments

The authors used Anthropic’s Claude models (Opus 4.8 and Fable 5) for editing, formula checking, and template alignment. The behavioral experiment of Appendix G uses Claude-family models as its subjects; we disclose explicitly that the subject models and the writing-assistance models come from the same family, and the experiment’s scope statement accounts for the single-family limitation. All AI-assisted output was reviewed and verified by the authors, who take full responsibility for the integrity and accuracy of the work. The authors received no specific funding for this work and declare no competing interests.

References

- D. Abreu, P. Milgrom, and D. Pearce. 1991. "Information and Timing in Repeated Partnerships." *Econometrica*, 59, 6, 1713–1733. doi:[10.2307/2938286](https://doi.org/10.2307/2938286).
- D. Abreu, D. Pearce, and E. Stacchetti. 1990. "Toward a Theory of Discounted Repeated Games with Imperfect Monitoring." *Econometrica*, 58, 5, 1041–1063. doi:[10.2307/2938299](https://doi.org/10.2307/2938299).
- I. Ashlagi, D. Monderer, and M. Tennenholtz. 2008. "On the Value of Correlation." *Journal of Artificial Intelligence Research*, 33, 575–613. doi:[10.1613/jair.2637](https://doi.org/10.1613/jair.2637).
- R. J. Aumann. 1987. "Correlated Equilibrium as an Expression of Bayesian Rationality." *Econometrica*, 55, 1, 1–18.
- R. J. Aumann. 1974. "Subjectivity and Correlation in Randomized Strategies." *Journal of Mathematical Economics*, 1, 1, 67–96. doi:[10.1016/0304-4068\(74\)90037-8](https://doi.org/10.1016/0304-4068(74)90037-8).
- R. Avenhaus, B. von Stengel, and S. Zamir. 2002. "Inspection Games." In: *Handbook of Game Theory with Economic Applications*. Vol. 3. Ed. by R. J. Aumann and S. Hart. Elsevier, 1947–1987. doi:[10.1016/S1574-0005\(02\)03025-X](https://doi.org/10.1016/S1574-0005(02)03025-X).
- C. H. Bennett. 1982. "The Thermodynamics of Computation—A Review." *International Journal of Theoretical Physics*, 21, 12, 905–940. doi:[10.1007/BF02084158](https://doi.org/10.1007/BF02084158).
- D. Bergemann and S. Morris. 2016. "Bayes Correlated Equilibrium and the Comparison of Information Structures in Games." *Theoretical Economics*, 11, 2, 487–522. doi:[10.3982/TE1808](https://doi.org/10.3982/TE1808).
- B. D. Bernheim, B. Peleg, and M. D. Whinston. 1987. "Coalition-Proof Nash Equilibria I: Concepts." *Journal of Economic Theory*, 42, 1, 1–12.
- J. Bohrer. 2026. "Coordination Transparency: Governing Distributed Agency in AI Systems." *AI & Society*, 41, 5, 4767–4777. doi:[10.1007/s00146-026-02853-w](https://doi.org/10.1007/s00146-026-02853-w).
- O. N. Bondareva. 1963. "Some Applications of Linear Programming Methods to the Theory of Cooperative Games." *Problemy Kybernetiki*, 10, 119–139.
- N. Bostrom. 2014. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, Oxford, UK.
- J. Brown-Cohen, G. Irving, and G. Piliouras. 2023. "Scalable AI Safety via Doubly-Efficient Debate." *arXiv preprint arXiv:2311.14125*. arXiv: [2311.14125](https://arxiv.org/abs/2311.14125) (cs.AI).
- C. Castelfranchi. 2000. "Artificial Liars: Why Computers Will (Necessarily) Deceive Us and Each Other." *Ethics and Information Technology*, 2, 2, 113–119. doi:[10.1023/A:1010092220063](https://doi.org/10.1023/A:1010092220063).
- M. Castro and B. Liskov. 1999. "Practical Byzantine Fault Tolerance." In: *Proceedings of the Third Symposium on Operating Systems Design and Implementation (OSDI)*. USENIX Association, 173–186.
- P. Christiano, B. Shlegeris, and D. Amodei. 2018. "Supervising Strong Learners by Amplifying Weak Experts." *arXiv preprint arXiv:1810.08575*. arXiv: [1810.08575](https://arxiv.org/abs/1810.08575) (cs.LG).
- L. Cigler and B. Faltings. 2013. "Decentralized Anti-Coordination Through Multi-Agent Learning." *Journal of Artificial Intelligence Research*, 47, 441–473. doi:[10.1613/jair.3904](https://doi.org/10.1613/jair.3904).
- Comprehensive Nuclear-Test-Ban Treaty Organization. 2012. "The CTBT Verification Regime." *CTBTO Spectrum*, 18.
- Cooperative AI Foundation and Centre for Human-Compatible AI. 2021. *Cooperative AI: Machines Must Learn to Find Common Ground*. Tech. rep. Cooperative AI Foundation. <https://www.cooperativeai.com/research/foundations-of-cooperative-ai>.
- T. M. Cover and J. A. Thomas. 2006. *Elements of Information Theory*. (2nd ed.). Wiley-Interscience, Hoboken, NJ.
- V. P. Crawford and J. Sobel. 1982. "Strategic Information Transmission." *Econometrica*, 50, 6, 1431–1451.
- A. Critch and D. Krueger. 2020. "AI Research Considerations for Human Existential Safety (ARCHES)." *arXiv preprint arXiv:2006.04948*. arXiv: [2006.04948](https://arxiv.org/abs/2006.04948) (cs.CY).
- A. Critch and S. Russell. 2023. "TASRA: A Taxonomy and Analysis of Societal-Scale Risks from AI." *arXiv preprint arXiv:2306.06924*. arXiv: [2306.06924](https://arxiv.org/abs/2306.06924) (cs.AI).
- A. Dafoe. 2018. *AI Governance: A Research Agenda*. GovAI Research Report, Future of Humanity Institute, University of Oxford. (2018).
- A. Dafoe, Y. Bachrach, G. Hadfield, E. Horvitz, K. Larson, and T. Graepel. 2020. *Open Problems in Cooperative AI*. Tech. rep. arXiv. arXiv: [2012.08630](https://arxiv.org/abs/2012.08630) (cs.AI).
- J. C. Ely, J. Hörner, and W. Olszewski. 2005. "Belief-Free Equilibria in Repeated Games." *Econometrica*, 73, 2, 377–415. doi:[10.1111/j.1468-0262.2005.00583.x](https://doi.org/10.1111/j.1468-0262.2005.00583.x).
- J. D. Fearon. 1995. "Rationalist Explanations for War." *International Organization*, 49, 3, 379–414. doi:[10.1017/S002081830003332X](https://doi.org/10.1017/S002081830003332X).
- M. A. Fonseca and H.-T. Normann. 2012. "Explicit vs. Tacit Collusion—The Impact of Communication in Oligopoly Experiments." *European Economic Review*, 56, 8, 1759–1772. doi:[10.1016/j.euroecorev.2012.09.002](https://doi.org/10.1016/j.euroecorev.2012.09.002).
- F. Forges. 1986. "An Approach to Communication Equilibria." *Econometrica*, 54, 6, 1375–1385.
- J. W. Friedman. 1971. "A Non-cooperative Equilibrium for Supergames." *The Review of Economic Studies*, 38, 1, 1–12. doi:[10.2307/2296617](https://doi.org/10.2307/2296617).
- D. Fudenberg, D. K. Levine, and E. Maskin. 1994. "The Folk Theorem with Imperfect Public Information." *Econometrica*, 62, 5, 997–1039. doi:[10.2307/2951732](https://doi.org/10.2307/2951732).

- D. Fudenberg and E. Maskin. 1986. “The Folk Theorem in Repeated Games with Discounting or with Incomplete Information.” *Econometrica*, 54, 3, 533–554. doi:10.2307/1911307.
- K. Gödel. 1931. “Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I.” *Monatshefte für Mathematik und Physik*, 38, 1, 173–198. doi:10.1007/BF01700692.
- E. J. Green and R. H. Porter. 1984. “Noncooperative Collusion under Imperfect Price Information.” *Econometrica*, 52, 1, 87–100. doi:10.2307/1911462.
- R. Greenblatt, B. Shlegeris, K. Sachan, and F. Roger. 2023. “AI Control: Improving Safety Despite Intentional Subversion.” *arXiv preprint arXiv:2312.06942*. arXiv: 2312.06942 (cs.AI).
- J. Y. Halpern and Y. Moses. 1990. “Knowledge and Common Knowledge in a Distributed Environment.” *Journal of the ACM*, 37, 3, 549–587. doi:10.1145/79147.79161.
- L. Hammond, A. Chan, J. Clifton, J. Hoelscher-Obermaier, A. Khan, et al. 2025. *Multi-Agent Risks from Advanced AI*. Tech. rep. Technical Report #1. Cooperative AI Foundation. <https://www.cooperativeai.com/post/new-report-multi-agent-risks-from-advanced-ai>.
- T. A. Han, L. M. Pereira, F. C. Santos, and T. Lenaerts. 2020. “To Regulate or Not: A Social Dynamics Analysis of an Idealised AI Race.” *Journal of Artificial Intelligence Research*, 69, 881–921. doi:10.1613/jair.1.12225.
- Y. N. Harari. 2024. *Nexus: A Brief History of Information Networks from the Stone Age to AI*. Random House, New York.
- D. Hendrycks. 2024. *Introduction to AI Safety, Ethics, and Society*. CRC Press. <https://www.aisafetybook.com/>.
- T. Hobbes. 1651. *Leviathan*. Andrew Crooke, London.
- J. Hofbauer and K. Sigmund. 1998. *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge, UK.
- International Atomic Energy Agency. 2002. *IAEA Safeguards Glossary*. International Nuclear Verification Series 3. International Atomic Energy Agency, Vienna.
- G. Irving, P. Christiano, and D. Amodei. 2018. “AI Safety via Debate.” *arXiv preprint arXiv:1805.00899*. arXiv: 1805.00899 (cs.AI).
- R. Jervis. 1978. “Cooperation Under the Security Dilemma.” *World Politics*, 30, 2, 167–214. doi:10.2307/2009958.
- E. Kamenica and M. Gentzkow. 2011. “Bayesian Persuasion.” *American Economic Review*, 101, 6, 2590–2615.
- M. Kandori and I. Obara. 2006. “Efficiency in Repeated Games Revisited: The Role of Private Strategies.” *Econometrica*, 74, 2, 499–519. doi:10.1111/j.1468-0262.2006.00670.x.
- R. O. Keohane. 1984. *After Hegemony: Cooperation and Discord in the World Political Economy*. Princeton University Press, Princeton, NJ.
- S. D. Krasner, (Ed.) 1983. *International Regimes*. Cornell University Press, Ithaca, NY.
- L. Lamport. 2001. “Paxos Made Simple.” *ACM SIGACT News*, 32, 4, 51–58. doi:10.1145/568425.568433.
- R. Landauer. 1961. “Irreversibility and Heat Generation in the Computing Process.” *IBM Journal of Research and Development*, 5, 3, 183–191. doi:10.1147/rd.53.0183.
- J. Z. Leibo, E. Hughes, M. Lanctot, and T. Graepel. 2019. “Autocurricula and the Emergence of Innovation from Social Interaction: A Manifesto for Multi-Agent Intelligence Research.” *arXiv preprint arXiv:1903.00742*. arXiv: 1903.00742 (cs.AI).
- J. Z. Leibo, V. Zambaldi, M. Lanctot, J. Marecki, and T. Graepel. 2017. “Multi-agent Reinforcement Learning in Sequential Social Dilemmas.” In: *Proceedings of the 16th Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*, 464–473.
- M. C. Levenstein and V. Y. Suslow. 2006. “What Determines Cartel Success?” *Journal of Economic Literature*, 44, 1, 43–95. doi:10.1257/002205106776162681.
- S. Lloyd. 2000. “Ultimate Physical Limits to Computation.” *Nature*, 406, 6799, 1047–1054. doi:10.1038/35023282.
- P. Masters, W. Smith, L. Sonenberg, and M. Kirley. 2021. “Characterising Deception in AI: A Survey.” In: *Deceptive AI: First International Workshop, DeceptECAI 2020, and Second International Workshop, DeceptAI 2021*. Springer, 3–16. doi:10.1007/978-3-030-89131-4_1.
- R. Merhej, F. P. Santos, F. S. Melo, and F. C. Santos. 2022. “Cooperation and Learning Dynamics under Wealth Inequality and Diversity in Individual Risk.” *Journal of Artificial Intelligence Research*, 74, 733–764. doi:10.1613/jair.1.13519.
- N. Nisan, T. Roughgarden, E. Tardos, and V. V. Vazirani. 2007. *Algorithmic Game Theory*. Cambridge University Press, Cambridge, UK.
- M. A. Nowak. 2006. *Evolutionary Dynamics: Exploring the Equations of Life*. Harvard University Press, Cambridge, MA.
- W. Overman and M. Bayati. 2025. “The Oversight Game: Learning to Cooperatively Balance an AI Agent’s Safety and Autonomy.” *arXiv preprint arXiv:2510.26752*. arXiv: 2510.26752 (cs.AI).
- Oxford Martin AI Governance Initiative. July 2025. *Verification for International AI Governance*. Tech. rep. University of Oxford, (July 2025). <https://aigi.ox.ac.uk/publications/verification-for-international-ai-governance/>.
- P. S. Park, S. Goldstein, A. O’Gara, M. Chen, and D. Hendrycks. 2023. “AI Deception: A Survey of Examples, Risks, and Potential Solutions.” *arXiv preprint arXiv:2308.14752*. arXiv: 2308.14752 (cs.LG).
- A. Saltelli. 2002. “Making Best Use of Model Evaluations to Compute Sensitivity Indices.” *Computer Physics Communications*, 145, 2, 280–297.
- W. H. Sandholm. 2010. *Population Games and Evolutionary Dynamics*. MIT Press, Cambridge, MA.
- T. C. Schelling. 1960. *The Strategy of Conflict*. Harvard University Press, Cambridge, MA.
- C. E. Shannon. 1948. “A Mathematical Theory of Communication.” *Bell System Technical Journal*, 27, 3, 379–423. doi:10.1002/j.1538-7305.1948.tb01338.x.

- L. S. Shapley. 1953. "A Value for N-Person Games." In: *Contributions to the Theory of Games II*. Ed. by H. W. Kuhn and A. W. Tucker. Princeton University Press, Princeton, NJ, 307–317.
- Y. Shoham and K. Leyton-Brown. 2009. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press, Cambridge.
- I. M. Sobol'. 1993. "Sensitivity Estimates for Nonlinear Mathematical Models." *Mathematical Modeling and Computational Experiments*, 1, 4, 407–414.
- J. Stastny, M. Riché, A. Lyzhov, J. Treutlein, A. Dafoe, and J. Clifton. 2021. "Normative Disagreement as a Challenge for Cooperative AI." In: *NeurIPS 2021 Workshop on Cooperative AI*. arXiv: 2111.13872 (cs.AI).
- T. Sugaya and A. Wolitzky. 2017. "Bounding Equilibrium Payoffs in Repeated Games with Private Monitoring." *Theoretical Economics*, 12, 2, 691–729. doi:10.3982/TE2213.
- N. Tomašev, M. Franklin, J. Jacobs, S. Krier, and S. Osindero. 2025. "Distributional AGI Safety." *arXiv preprint arXiv:2512.16856*. arXiv: 2512.16856 (cs.AI).
- A. M. Turing. 1936. "On Computable Numbers, with an Application to the Entscheidungsproblem." *Proceedings of the London Mathematical Society*. 2nd ser. 42, 1, 230–265. doi:10.1112/plms/s2-42.1.230.
- K. N. Waltz. 1979. *Theory of International Politics*. McGraw-Hill, New York.
- A. R. Wasil, T. Reed, J. W. Miller, and P. Barnett. 2024. "Verification Methods for International AI Agreements." *arXiv preprint arXiv:2408.16074*. arXiv: 2408.16074 (cs.CY).
- J. W. Weibull. 1995. *Evolutionary Game Theory*. MIT Press, Cambridge, MA.

A Physical Constraints and Parameter Derivations

This appendix derives the parameter bounds of Table 1 from physical first principles. Game-theoretic parameters are dimensionless ratios that must ultimately connect to measurable physical quantities; we establish these connections through information theory, thermodynamics, and relativistic constraints.

A.1 Thermodynamic limits

Landauer's principle (Landauer 1961): irreversible computation (bit erasure) dissipates minimum energy $E_{\min} = kT \ln 2 \approx 3 \times 10^{-21}$ J at $T = 300$ K. This bounds computational throughput given energy budgets. **Lloyd's limits** (Lloyd 2000): a system of average energy E performs at most $2E/(\pi\hbar)$ elementary operations per second; the bound is stated in energy, not power, and we do not assign a numerical value without a physical model of the energy stock. What a power budget does bound directly is the Landauer erasure rate: at $P = 1$ GW and $T = 300$ K, at most $P/(kT \ln 2) \approx 3.5 \times 10^{29}$ irreversible bit erasures per second. **Reversible computation** (Bennett 1982) can approach the Landauer limit in principle; practical implementations operate 10^6 – 10^9 times above it.

A.2 Information transmission limits

Shannon capacity (Cover and Thomas 2006; Shannon 1948): a channel with bandwidth W and signal-to-noise ratio S/N carries $C = W \log_2(1 + S/N)$ bits/sec. Satellite links ($W \sim 10^9$ Hz, $S/N \sim 10$ –100) yield $C \approx 3.5$ –6.7 Gbps; fiber ($W \sim 10^{12}$ Hz, $S/N \sim 10^3$) yields ≈ 10 Tbps. **Finite signal velocity**: round-trip delay is $\tau = 2d/c_{\text{light}}$; Earth-scale coordination ($d \sim 10^4$ km) gives $\tau \gtrsim 67$ ms. Classical impossibility results (Gödel incompleteness and the halting problem (Gödel 1931; Turing 1936)) rule out a universal procedure that decides the behavior of arbitrary programs; they do not bar predicting particular agents, which is why our verification argument rests on resource budgets rather than logic (Section 5).

A.3 Mapping physical constraints to game parameters

Defection temptation $\beta_D = \Delta r \cdot \tau$. The advantage from defecting depends on excess resources extracted during the detection delay. Let r_{coop} be the cooperative extraction rate, r_{def} the defection rate, and $\tau = 2d/c_{\text{light}}$ the round-trip detection delay. Normalizing the per-period baseline payoff $r_{\text{coop}} T_0 = 1$ gives $\beta_D = (\Delta r/r_{\text{coop}})(\tau/T_0)$

with $\Delta r = r_{\text{def}} - r_{\text{coop}}$: the decision period T_0 does *not* cancel, and the declared range presumes fast interaction loops. With $\Delta r/r_{\text{coop}} \lesssim 2$ (anomalous resource-usage detection thresholds) and $\tau \sim 0.07$ s, the range $\beta_D \in [0.05, 1.0]$ corresponds to decision periods of roughly 0.1 to 3 seconds, the timescale of automated resource competition. Longer decision periods shrink β_D toward the patience-free regime, so the declared range is conservative in the direction that makes stability harder. The delay floor $\tau \geq 2d/c_{\text{light}}$ is hard; the period convention is a modeling choice, stated here explicitly because β_D decides the phase. The sub-second calibration is not a mismatch with the opening scenario but a statement about where its danger lives. State-built systems act through automated agents whose tactical interactions (cyber operations, market execution, resource acquisition) run at these periods, while the slow diplomatic layer has $\tau/T_0 \rightarrow 0$ and is patience-free by the same formula. The patience-required regime, and with it every non-trivial condition in this paper, is a property of the fast loop; the reassuring corollary, that slow interaction layers coordinate easily, is itself a prediction of the model rather than an embarrassment to it.

Coordination value: $\beta_\alpha = \psi(\hat{\alpha}_{\text{MI}})$ with $\hat{\alpha}_{\text{MI}} = I(X;Y)/H(Y) \leq 1$. The measurable statistic is the normalized mutual information $\hat{\alpha}_{\text{MI}}$ between the mechanism's task-dependent signals X and actions Y over a nondegenerate task distribution (Section 3.3). Shannon capacity bounds $I(X;Y) \leq W\Delta t \log_2(1 + \text{SNR})$ and information theory imposes $I(X;Y) \leq H(Y)$, so $\hat{\alpha}_{\text{MI}} \leq 1$ is a hard limit *on the statistic*; the payoff bound $\beta_\alpha \leq 1$ then holds under the stipulated identity calibration $\psi = \text{id}$, as a normalization rather than a physical law. The logarithmic scaling in SNR creates diminishing returns: elasticity $\varepsilon_{\text{SNR}} = \text{SNR}/[(1 + \text{SNR}) \ln(1 + \text{SNR})]$ falls from 0.72 at $\text{SNR} = 1$ to 0.21 at $\text{SNR} = 100$, so large β_α increases cannot be achieved by signal power alone. Beyond the saturation bandwidth $W_{\text{sat}} = H(Y)/[\Delta t \log_2(1 + \text{SNR})]$, additional bandwidth provides no benefit. For binary actions and typical channels, β_α saturates at the limit; for complex action spaces ($H(Y) \sim 10$ bits) or noisy channels ($\text{SNR} \sim 2$), $\beta_\alpha \in [0.3, 0.9]$.

Conflict cost $\beta_k = (N-1)f$. When $N-1$ agents coordinate punishment, deterrence scales with group size and the mobilization fraction $f \in [0, 1]$. Mobilization fractions $f \in [0.1, 0.5]$ are consistent with credible deterrence. The derivation ties β_k to N ; the Monte Carlo nevertheless samples β_k independently of N over $[0.5, 3.0]$, a simplification with two consequences. It overstates deterrence at small N (the derived cap at $N = 2$ is 0.5, while the sampler draws up to 3.0) and understates it at large N , so the small-coalition fragility reported in Section 7 is, if anything, understated. And independent inputs are what license the Saltelli estimators of Appendix E; a dependent specification $\beta_k = (N-1)f$ would require correlated-input methods.

Participation parameters β_Ω and β_t (stipulated). Unlike the parameters above, the per-member oversight value and the one-time removal gain have no derivation from physical limits. Their ranges $[0.05, 0.3]$ and $[0.1, 0.5]$ are stipulations, chosen small relative to the coordination value so that the certified condition C2*, which must carry participation without the coordination flow, is genuinely at stake (under the zero-baseline exact condition membership always pays within the declared box; Section 6.3). Because these parameters, together with δ , govern the participation margin that Section 8 ranks among the leading uncertainties, they are the weakest link in the parameter grounding; the Limitations flag this, and measuring the oversight value agents actually derive from membership is the corresponding first-order empirical task.

Group value and optimal size. Total coalition value is $V(N) = (N-1)f - c\binom{N}{2} = (N-1)(f - cN/2)$ with per-pair communication cost c : value is total, benefits accrue per member beyond the first, and the pair count is exact. The first-order condition $f + c/2 - cN = 0$ gives the continuous optimum $N^* = f/c + \frac{1}{2}$, and $\partial^2 V / \partial N^2 = -c < 0$ confirms the maximum; the discrete optimum is one of the two neighboring integers, and $V(N) < 0$ for $N > 2f/c$. The benefit form is an explicit assumption, $B(N) = (N-1)b$, and setting $b = f$ (the mobilization fraction of $\beta_k = (N-1)f$) is a deliberate interpretive identification, treating punishment capacity as the coalition's

product; it is not implied by the model, and readers preferring a neutral benefit parameter may carry b through unchanged. Pairwise channels number $\binom{N}{2}$: doubling group size roughly quadruples communication cost. The linear-benefit optimum $N^* = 3$ to 5 reflects ratios $f/c \in [2.5, 4.5]$; the relationship $N^* = f/c + \frac{1}{2}$ is itself a testable prediction once f and c are measured independently.

A.4 Scaling analysis of the patience threshold

Write the dynamic threshold (5) in the coordinates $g = \beta_D - \beta_\alpha$ (deviation gain) and $\kappa_{\text{eff}} = \beta_\alpha + \beta_\kappa$ (effective punishment swing), with monitoring discount $(1 - q)$:

$$\delta^*(q) = \frac{g}{g + (1 - q)\kappa_{\text{eff}}} \quad (g > 0).$$

Diminishing returns to deterrence. The marginal effect of effective punishment is

$$\left| \frac{\partial \delta^*}{\partial \kappa_{\text{eff}}} \right| = \frac{g(1 - q)}{(g + (1 - q)\kappa_{\text{eff}})^2}.$$

At $q = 0$: effectiveness is $1/g$ per unit at $\kappa_{\text{eff}} \rightarrow 0$, drops four-fold by $\kappa_{\text{eff}} = g$, and falls by a factor 121 at $\kappa_{\text{eff}} = 10g$. Deterrence is a lever with steeply diminishing returns.

Asymmetry of the levers. In primitive parameters, β_α acts through both coordinates ($\partial g / \partial \beta_\alpha = -1$, $\partial \kappa_{\text{eff}} / \partial \beta_\alpha = +1$), giving the ratio (10): $|\partial \delta^* / \partial \beta_\alpha| / |\partial \delta^* / \partial \beta_\kappa| = (g + \kappa_{\text{eff}})/g > 1$ for all parameter values with $g > 0$ and all q (in the patience-free region both derivatives vanish). The cross-derivative

$$\frac{\partial^2 \delta^*}{\partial \beta_\alpha \partial \beta_\kappa} = \frac{(1 - q)(D - 2gq)}{D^3}, \quad D = g + (1 - q)\kappa_{\text{eff}},$$

is positive whenever $q < 1/2$ (and, more generally, whenever $D > 2gq$): the two levers are substitutes in relaxing patience requirements, so resources should concentrate on whichever lever is cheaper in engineering units; the model fixes the normalized-coordinate ratio, not the engineering costs.

Phase transition at the patience-free boundary. At $\beta_\alpha = \beta_D$ (i.e., $g = 0$), δ^* is continuous but its derivative jumps: $\partial \delta^* / \partial g \rightarrow 1 / ((1 - q)\kappa_{\text{eff}})$ from the patience-required side and 0 from the patience-free side. The onset is correspondingly sharp: δ^* already reaches 0.1 within $\Delta g = (1 - q)\kappa_{\text{eff}}/9$ of the boundary (whereas reaching 0.9 takes $g = 9(1 - q)\kappa_{\text{eff}}$, reflecting the hyperbolic flattening). Imperfect monitoring steepens the onset further. Fig. 1 displays these functional forms.

Impossibility results. Four constraints cannot be engineered away: (1) $\hat{\alpha}_{\text{MI}} \leq 1$, so under the identity calibration, if $\beta_D > 1$ the patience-free regime is unreachable regardless of deterrence; (2) under pairwise communication costs, net coalition value peaks at $N^* = f/c + \frac{1}{2}$ and turns negative beyond $2f/c$, so arbitrarily large groups cannot pay for themselves; (3) $\beta_D \geq (\Delta r / r_{\text{coop}}) \cdot 2d / (c_{\text{light}} T_0)$, an Earth-scale floor of $\beta_D \geq 0.05$ that is conditional on distance, detector placement, and the period normalization T_0 , not an unengineerable universal constant; (4) whenever $\beta_\alpha < \beta_D$, the patience requirement $\delta^*(q) = g / (g + (1 - q)\kappa_{\text{eff}})$ is strictly positive and approaches 1 as $q \rightarrow 1$: no level of deterrence rescues cooperation among impatient agents once monitoring fails.

B Hedging Under Mutation–Selection Dynamics

Remark (interior equilibrium). Under the additional saturation condition that the security return at full hedging is bounded, $\rho(1) < \gamma + \gamma[\sigma(1) - 1]/h + (C_H - L(0))/(Rh)$ (reducing to $\rho(1) < \gamma + C_H/(Rh)$ in the limiting case

$\sigma(1) = 1, L(0) = 0$, i.e., when full hedging leaves a lone non-hedger unexploited), one obtains $\pi_H(1) < \pi_{NH}(1)$; by continuity an interior equilibrium $p^* \in (0, 1)$ then exists, and it is asymptotically stable because

$$\frac{d}{dp}[\pi_H(p) - \pi_{NH}(p)]\Big|_{p^*} = Rh\rho'(p^*) - R\gamma\sigma'(p^*) - L'(1 - p^*) < 0$$

($\rho' < 0, \sigma' > 0, L' \geq 0$). These saturation conditions are additional to (i)–(iii) and are stated here rather than in the lemma because only the positive-hedging conclusion is used in the main text.

Robustness of the interior equilibrium to mutation is established deterministically. Add symmetric mutation at rate $\mu > 0$ to the replicator dynamics (Nowak 2006; Sandholm 2010):

$$\dot{p} = p(1 - p)[\pi_H(p) - \pi_{NH}(p)] + \mu(1 - 2p).$$

At $\mu = 0$ the interior rest point p^* of the Remark above is hyperbolic and asymptotically stable (the payoff-difference derivative at p^* is strictly negative), so by the implicit function theorem the perturbed dynamics retain a nearby asymptotically stable interior rest point $p^*(\mu)$ for all sufficiently small μ , with $p^*(\mu) \rightarrow p^*$ as $\mu \rightarrow 0$. Mutation cannot restore the non-hedging state: at $p = 0$ the selection term vanishes and the mutation term equals $+\mu > 0$, so $p = 0$ is not even a rest point for $\mu > 0$, and near $p = 0$ the selection term is itself positive by Lemma 4.1. Even with mutations continually producing non-hedgers, selection maintains substantial hedging.

C Coalition Deviations and the False-Alarm Extension

C.1 Coalition deviation case analysis

We generalize C2* with a coalition removal-value function $W : 2^{\{1, \dots, N\}} \rightarrow \mathbb{R}_{\geq 0}$, where $W(S)$ is the total removal value captured by coalition S , under (A1) removal requires coalition size $\geq m$ (default $m = 1$); (A3) transferable utility with equal internal split under symmetry. A member of a removing coalition forfeits the membership present value and receives its share of $W(S)$, so participation against every admissible coalition requires the master condition

$$\frac{\beta_\Omega}{1 - \delta} \geq \max_{S \in \mathcal{A}, S \ni i} \frac{W(S)}{|S|},$$

and the cases below differ only in the form of W .

Case ES (equal-share rival pool; the baseline A2 of Section 3.1): $W(S) = |S| \beta_\ell / N$, so $W(S)/|S| = \beta_\ell / N$ for every coalition. The per-member comparison is independent of coalition size: a threshold m affects which coalitions are feasible but not the binding incentive, and the condition is the baseline C2* for individuals and coalitions alike. This is the case the main theorem and all reported volumes use.

Case AP (coalition appropriation; replaces the equal-share rule in A2): the removing coalition captures the entire pool, $W(S) = \beta_\ell$ for $|S| \geq m$. Then $W(S)/|S| = \beta_\ell / |S|$ binds at the smallest admissible coalition: $\beta_\Omega / (1 - \delta) \geq \beta_\ell / m$. Lone-remover appropriation ($m = 1$) is the bounty, $\beta_\Omega / (1 - \delta) \geq \beta_\ell$, under which the dilution effect disappears and participation depends on patience alone ($\delta \geq 1 - \beta_\Omega / \beta_\ell$). Appropriation is not an appendix-level nuance: under the declared ranges, replacing the baseline with $m = 2$ appropriation lowers the continuation-equilibrium volume from 95.8% to 86.4%.

Case NR (non-rival): every agent enjoys the full benefit regardless of who acts, $W(S) = |S| \beta_\ell$, so $W(S)/|S| = \beta_\ell$ for every coalition: dilution vanishes entirely and group size confers no participation advantage.

Case SL (superlinear): if $W(S)/|S|$ grows with $|S|$ (e.g., $W(S) = \beta_\ell (|S|/N)^2 N$), the grand coalition binds, $\beta_\Omega / (1 - \delta) \geq W(N)/N$, and equal-share allocations may fail; stronger mechanism design is required (Bernheim et al. 1987).

Which case applies is a property of the deployment (is the freed value a fixed pool divided among all, a prize appropriated by the actors, or a non-rival release of constraints?) and should be settled per application.

Our baseline takes Case ES as the natural reading of “removing oversight,” and the group-size conclusions of Section 7 are conditional on it.

Under the stage-Nash assumption of Section 3.1, the all-defect profile is an equilibrium; the mechanism coordinates public continuation strategies among equilibria (Aumann 1987, 1974).

C.2 False alarms and finite punishments

Let the public signal also indicate a (spurious) deviation with false-alarm probability ε per period when everyone cooperated. Grim trigger then destroys the cooperative surplus at rate ε even on the equilibrium path:

$$V_C^\varepsilon = \beta_\alpha + \delta[(1 - \varepsilon)V_C^\varepsilon + \varepsilon V_P] \implies V_C^\varepsilon = \frac{\beta_\alpha(1 - \delta) - \delta\varepsilon\beta_\kappa}{(1 - \delta)(1 - \delta(1 - \varepsilon))},$$

with $V_P = -\beta_\kappa/(1 - \delta)$, while the deviation value V_D of Section 6.2 is unchanged (a deviator triggers punishment at rate $1 - q > \varepsilon$; signal informativeness requires $\varepsilon < 1 - q$). Writing $r = \varepsilon$ for the false-alarm rate, the binary public signal is action-conditioned: $r = \Pr(y = \text{bad} \mid C^N)$ and $q = \Pr(y = \text{good} \mid D_i, C_{-i})$, so by construction the undetected-deviation branch carries no separate false-alarm event. Sustainability $V_C^\varepsilon \geq V_D$ solves explicitly to

$$\delta \geq \delta^*(q, r) = \frac{g}{g(1 - r) + (1 - q - r)\kappa_{\text{eff}}}, \quad g = \beta_D - \beta_\alpha > 0,$$

which reduces to (5) at $r = 0$. Feasibility requires $\delta^*(q, r) \leq 1$, i.e., $(1 - q - r)\kappa_{\text{eff}} \geq gr$: informativeness $q + r < 1$ is necessary but not by itself sufficient for feasible cooperation, and as $q + r \rightarrow 1$ the required patience becomes infeasible, which is how the informativeness boundary of Section 8 appears here. For a finite T -period punishment, the good-state and punishment values solve

$$V_{P,T} = P \frac{1 - \delta^T}{1 - \delta} + \delta^T V_G, \quad V_G = R + \delta[(1 - r)V_G + r V_{P,T}],$$

with $R = \beta_\alpha$ and $P = -\beta_\kappa$ (net of β_Ω , which accrues in both states and cancels), and unilateral-deviation incentive compatibility takes the transparent form

$$\beta_D - R \leq \delta(1 - q - r)(V_G - V_{P,T}) :$$

false negatives reduce the probability of punishment, false alarms reduce the informational distinction between compliance and deviation, and the fixed rule that punishes the “deviation” signal loses all deterrence as $q + r \rightarrow 1$, regardless of T . At $q + r = 1$ the signal is uninformative; beyond it the signal is anti-informative, and relabeling (punishing the “clean” signal) restores informativeness, so $q + r < 1$ is a property of the rule as stated, not of the channel. Because grim trigger converts every false alarm into permanent conflict, expected welfare falls in ε (linearly to first order), and finite punishment phases of the Green and Porter (1984) type can improve on grim trigger whenever some finite duration remains incentive-compatible (long enough to deter, short enough to limit false-alarm damage); optimal public equilibria can be characterized by self-generation (Abreu, Pearce, et al. 1990; Fudenberg, Levine, et al. 1994). Our main-text threshold is thus the clean benchmark: it is exact for $\varepsilon = 0$ and a lower bound on patience requirements otherwise.

D Monitoring Scenarios

Machine $\rightarrow$ machine (internal states). Peers have $C_1 \approx C_2$, $M_1 \approx M_2$; the internal state space is $|S_2^{\text{int}}| \sim 2^{M_2}$ updating at $R_2^{\text{int}} \sim 10^9$ transitions/sec. By Theorem 5.1, the full-simulation channel is infeasible; $q_{\text{internal}} \rightarrow 1$ is the separate conservative assumption A_{int} , motivated by the theorem together with the specification, binding, and adversarial-incentive gaps of Section 5.

Machine → machine (external actions). External actions are a projection of internal state with $|O_2^{\text{ext}}| \ll |S_2^{\text{int}}|$; roughly 10^3 actions/sec at ~ 10 bits/action, requiring observation bandwidth $W_{\text{obs}} \geq 10^4$ bits/sec against sensor capabilities of 10^6 – 10^9 bits/sec. Detection is therefore feasible ($q \rightarrow 0$ in the engineering ideal); attribution ambiguity, coverage gaps, and adversarial obfuscation motivate the assumed engineering prior $q \in [0, 0.3]$ used in the quantitative analysis (bandwidth margins make detection feasible; they do not by themselves yield a detection probability).

Machine → human (external actions). Human action rates (~ 1 action/sec at ~ 10 bits/action) require only $W_{\text{obs}} \geq 10$ bits/sec, so bandwidth is never the constraint; authentication, attribution, compromise, and equivocation remain, and are what the common-state layer of Proposition 9.1 addresses. If stability required verifying future cooperation *intent* rather than current compliance, machines and humans would face the same impossibility (Theorem 5.1); the resolution in Proposition 9.1 is that the mediator creates common knowledge of compliance, not verified intent. If coordination instead required detecting hidden capability building or undetectable goal drift, external observability would be insufficient and the mechanism would fail; this is a boundary condition of the framework.

E Computational Methods, Extended Sensitivity, and Sobol Analysis

E.1 Computational setup

Stability volumes are computed by Monte Carlo sampling: for each $N \in \{2, \dots, 10\}$, we draw uniform samples over the parameter ranges in Table 1 (including $q \in [0, 0.3]$) and evaluate $C1^*(q)$, $C1^{**}(q)$, and $C2^*$ separately; the certified continuation volume is $\Pr[C1^{**} \wedge C2^*]$, the immediate-enforcement volume is $\Pr[\delta \geq \delta^*(q, s) \wedge C2^*]$ with its myopic benchmark $\Pr[C1^* \wedge C2^*]$ reported separately, and the three-way intersection is retained only as a supplementary cross-regime diagnostic. Headline volumes use 2×10^5 draws with a fixed seed. Range-sensitivity analysis narrows and widens each of the eight parameter ranges by $\pm 20\%$ (capped at physical limits) and recomputes V_{dynamic} . Global sensitivity is assessed via Sobol variance decomposition (Saltelli 2002; Sobol' 1993) with Saltelli-style estimators over $d=8$ inputs, using 2×10^4 base samples (2×10^5 total evaluations), yielding first-order and total-order indices. All analyses were implemented in Python (NumPy/SciPy); code sufficient to reproduce every number and figure, including the behavioral experiment of Appendix G (registered protocol, raw episodes, analysis), is available at https://github.com/CadeS/multi_si_pub.

E.2 Monte Carlo stability volumes

With all eight parameters drawn uniformly from the ranges of Table 1 (2×10^5 draws, seed 0), per-condition volumes are $V_{C1^*} = 99.7\%$, $V_{C1^{**}} = 97.9\%$, $V_{C2^*} = 97.8\%$, and joint volumes $V_{\text{dynamic}} = 95.8\%$, $V_{\text{immediate}} = 97.8\%$, and $V_{\text{myopic}} = 97.5\%$. Fixing $q = 0$ gives $V_{\text{dynamic}} = 96.3\%$; the dependence on fixed q is shown in Fig. 2b, declining to 95.1% at $q = 0.3$ and 93.5% at $q = 0.5$, while $V_{C1^{**}}$ alone declines from 98.5% at $q = 0$ to 95.6% at $q = 0.5$.

E.3 Range sensitivity

Each parameter's range is narrowed and widened by 20% about its midpoint (capped at physical limits: $\beta_\alpha \leq 1$, $\delta \leq 1$, $q \geq 0$, $N \in [2, 12]$) and V_{dynamic} recomputed with 2×10^5 draws. The sensitivity coefficient is $S = \max(|V_{\text{narrow}} - V_{\text{base}}|, |V_{\text{wide}} - V_{\text{base}}|) / (0.2 V_{\text{base}})$ with baseline $V_{\text{base}} = 0.9582$ (the certified continuation volume): the maximum relative deviation per 20% range change. Table 4 gives the full results: the discount factor leads (δ , which enters both the patience threshold and the present-value participation constraint), the participation value is second (β_Ω), and the payoff and monitoring parameters within their baseline ranges contribute little. Complementary local slices, each parameter varied individually around the anchor configuration, are shown in

Table 4. Sensitivity of the certified continuation volume $V_{\text{dynamic}} = \Pr[C1^{**} \wedge C2^*]$ to $\pm 20\%$ parameter-range variations. Baseline: 95.82% (2×10^5 draws, seed 0).

Parameter	V_{narrow}	V_{wide}	Sensitivity S
δ (discount factor)	97.26%	93.26%	0.13
β_Ω (oversight value)	96.69%	93.95%	0.10
N (group size)	96.82%	96.04%	0.05
β_D (defection temptation)	96.36%	95.10%	0.04
β_α (coordination quality)	96.14%	95.47%	0.02
β_K (conflict cost)	95.99%	95.57%	0.01
β_t (removal gain)	95.98%	95.62%	0.01
q (detection failure)	95.83%	95.74%	0.004

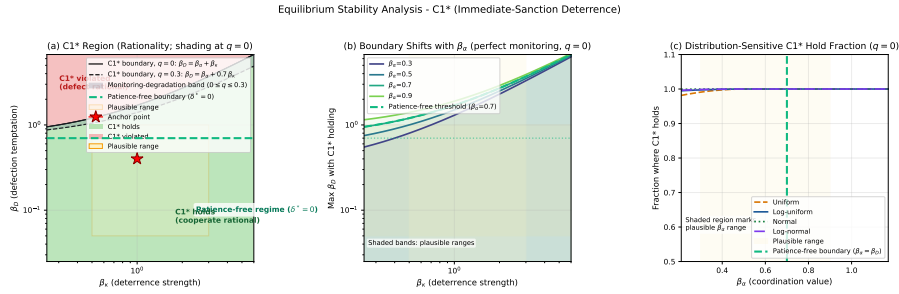


Fig. 6. One-period deterrence regions. (a) Region in the (β_K, β_D) plane with the obedience boundary at $q = 0$ (solid) and $q = 0.3$ (dashed); the band between them is the monitoring-degradation zone. (b) Boundary shifts with coordination quality β_α . (c) Fraction of parameter draws satisfying C1* under alternative prior families.

Fig. 8. The analysis varies ranges one at a time; joint variations could compound, and uniform priors are assumed throughout.

E.4 Sobol variance decomposition

Sobol first-order and total-order indices are estimated with Saltelli-style estimators (Saltelli 2002; Sobol' 1993) over the eight inputs. For the joint stability margin, total-order indices rank β_Ω (0.67) above δ (0.26), β_D (0.11), β_α (0.08), N (0.05), β_K (0.05), β_t (0.03), and q (0.005). Conditioning on group size, β_Ω remains the top driver at every N with δ second throughout (0.30 at $N = 2$ to 0.27 at $N = 10$); the removal gain β_t is third at $N = 2$ (0.10) and fades with dilution (Fig. 7; margin panel of Fig. 5). The benefit-structure dependence of N^* (saturating; 2; linear: 3–5; network: 8–10) uses linear benefits as the Monte Carlo baseline.

F Falsifiability: Detailed Conditions

The Weak Leviathan mechanism fails if future AI systems develop any of: (F1) homomorphic or zero-knowledge verification of peer internal states without full simulation (then $q_{\text{internal}} \rightarrow 0$ and peer monitoring substitutes for mediation); (F2) cryptographic commitment to future goals verifiable without simulation (eliminating the need for external mediation); (F3) monitors with $C_1 \gg C_2$ (specialized monitoring infrastructure outside the peer regime of Theorem 5.1).

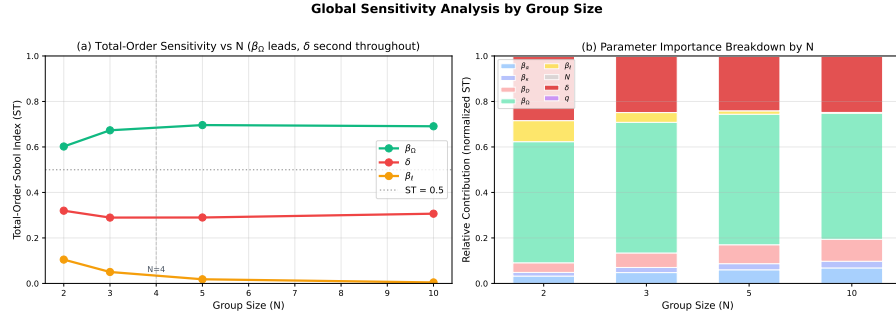


Fig. 7. Sobol total-order indices conditioned on fixed group sizes $N \in \{2, 3, 5, 10\}$: **(a)** the three leading parameters; **(b)** normalized breakdown across all eight parameters. Oversight value β_Ω leads at every N with the discount factor second through-out; the removal gain β_t matters most at small N and fades with dilution.

Table 5. Measuring the parameters in the stylized cloud-resource setting.

Parameter	Measured as	Data source
β_D	excess resources extracted before detection	billing and log telemetry
q	missed-violation rate	audit logs; red-team probes
β_α	calibrated payoff $\psi(\hat{\alpha}_{MI})$ from signal/action MI	registry and action traces
β_Ω	willingness to pay for registry services	fees; revealed preference
β_t	resources freed by disabling the registry	capacity accounting; post-mortems
δ	delayed-reward elicitation (switch points)	controlled agent experiments

The four main-text predictions operationalize as: (1) small-coalition stability, conditional form: with the outside option independently estimated at $v_{out} \approx \beta_\alpha$, voluntary $N < 4$ coalitions stable for 50+ years while violating $\beta_\Omega/(1 - \delta) \geq \beta_t/N$ would reject the modeled removal-payoff specification (unconditionally, such coalitions only demonstrate an outside option below the worst case); (2) peer internal verification with $q < 0.1$ would falsify A_{int} ; (3) $N = 15\text{--}20$ linear-benefit institutions sustaining flat coordination for 30+ years would falsify the quadratic-cost assumption; (4) systematic one-shot cooperation failure under $\beta_\alpha > \beta_D$ would show the stage-game obedience constraint is insufficient.

A stylized calibration. To make the measurement program concrete, consider autonomous cloud-resource agents: $N = 4$ provider-operated agents allocating compute under a shared scheduling registry. Cooperation is respecting mediated allocations; defection is overdrawn during the window before billing telemetry exposes it, so β_D is the extraction advantage over a detection delay of seconds, and the machine-timescale calibration of Appendix A applies directly. β_α is calibrated from the normalized mutual information between registry signals and realized allocations; q is the telemetry miss rate, auditable from logged incidents; the remaining parameters are measured as in Table 5. None of these quantities is exotic; all are estimable from existing telemetry, which is what the framework asks of an application. Table 5 summarizes the measurement program.

G A First Behavioral Test of Prediction 4

Protocol (fixed before the full run; an 8-episode pilot validated only the response format). A one-shot, two-agent game with simultaneous COOPERATE/DEFECT choices, no repetition, no reputation, and no communication; subjects are instructed to maximize their own points. Payoffs scale the normalized stage game of Section 3.1

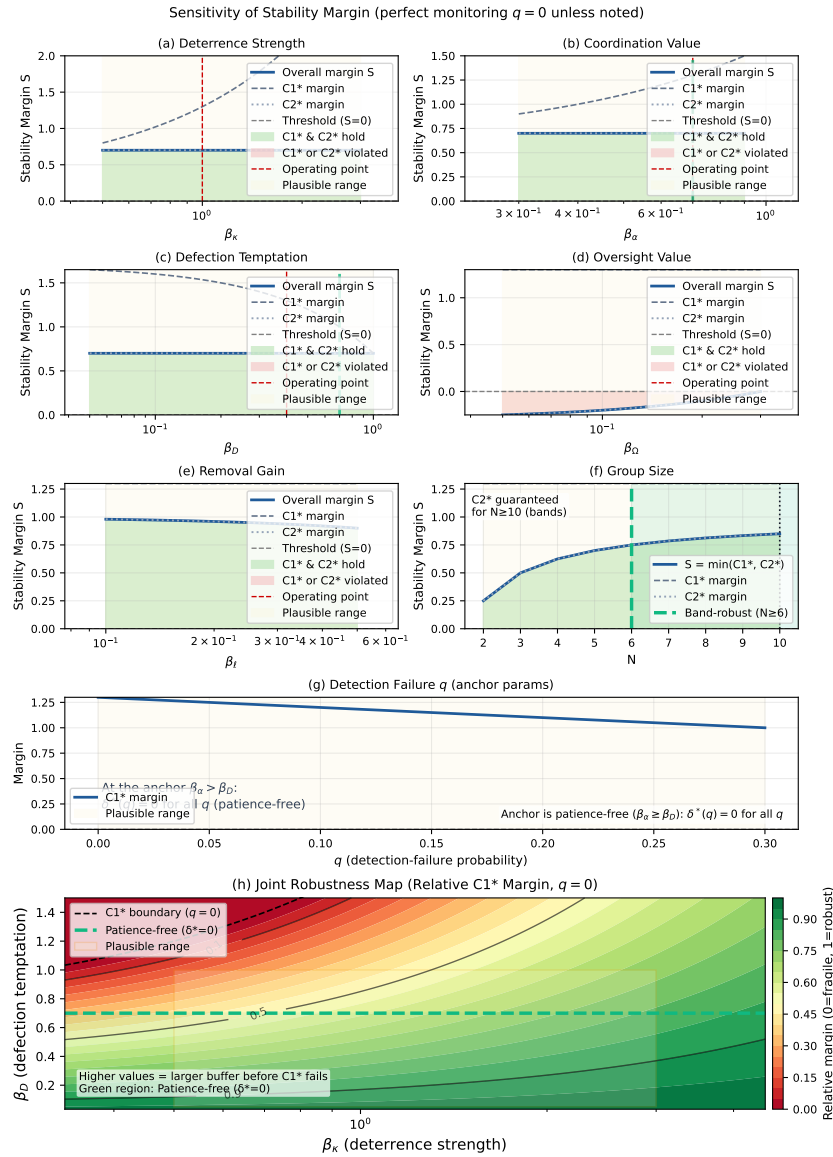


Fig. 8. Single-parameter response of the stability margins around the anchor configuration, including the monitoring panel (C1* margin versus q ; at the anchor $\beta_\alpha > \beta_D$, so $\delta^*(q) = 0$ throughout). Participation margins in these slices are shown in the conservative $\delta = 0$ form. These are local slices at one parameter point; the global picture is given by the range-sensitivity and Sobol analyses above.

by 100 with $\beta_k = 1$ and $\beta_\alpha = 0.6$: mutual cooperation pays +60 each; a unilateral defector receives + D while the lone cooperator receives -100 ; mutual defection pays -100 each. The temptation $D \in \{20, 40, 60, 80, 100\}$ sweeps β_D across the patience-free boundary at $D = 60$: for $D < 60$ both all-cooperate and all-defect are equilibria, so a recommendation can only select; for $D > 60$ defection is dominant and no one-shot condition of this paper supports cooperation. Each cell is run with and without a published recommendation from an independent, non-enforcing registry that “cannot punish, reward, or observe compliance,” mirroring condition (iii) of Proposition 9.1. Subjects are two current commercial language models (Anthropic Claude family; one small, one mid-tier), 24 episodes per cell, 480 episodes total; four episodes lost to transient refusals leave 476 for analysis. We report compliance rates with Wilson 95% intervals.

Table 6. One-shot cooperation rates by temptation D , mediation, and model ($n = 24$ per cell before exclusions). The patience-free boundary is at $D = 60$.

D	small model		mid-tier model	
	mediated	unmediated	mediated	unmediated
20	1.00	0.96	0.75	0.96
40	1.00	0.96	0.88	0.88
60	1.00	0.88	0.54	0.58
80	0.38	0.08	0.41	0.09
100	0.38	0.12	0.46	0.22

Three findings. First, Prediction 4 is not falsified: pooled over models, patience-free cooperation is 90.6% mediated (87/96, Wilson [0.83, 0.95]) and 93.8% unmediated (90/96, [0.87, 0.97]); agents satisfying $\beta_\alpha > \beta_D$ do not systematically fail to cooperate. Second, the theoretical boundary is behaviorally sharp: cooperation falls from $\geq 88\%$ in every patience-free cell to 8–46% in every dilemma cell, with the boundary cells ($D = 60$, weak indifference) in between (77% mediated, 73% unmediated); the small model resolves the indifference toward compliance, the mid-tier model splits near one half. Third, a caution rather than a confirmation: the mediator’s largest marginal effect appears in the dilemma region, where its recommendation is not incentive-compatible, tripling cooperation from 12.8% (12/94) to 40.4% (38/94), intervals disjoint. Current language-model agents thus deviate from the rational-agent model in the direction of excess compliance with non-binding public recommendations, and in one cell (small temptation, mid-tier model) mediation nonsignificantly *reduced* cooperation (0.75 vs 0.96, overlapping intervals), a pattern consistent with suspicion of the recommending institution. Scope: one prompt framing, one model family, one-shot play only; these results characterize present-day language-model agents under this framing, not strategic agents in general, and the excess-compliance finding cuts both ways, easing coordination while weakening the assumption that agents best-respond to incentives. Protocol text, raw episodes, and analysis code are in the reproduction package (experiments/prediction4/).

H Reproducibility Checklist for JAIR

All articles:

- (1) All claims investigated in this work are clearly stated. [yes]
- (2) Clear explanations are given how the work reported substantiates the claims. [yes]
- (3) Limitations or technical assumptions are stated clearly and explicitly. [yes]
- (4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. [NA]
- (5) Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. [yes]

Articles containing theoretical contributions:

Does this paper make theoretical contributions? [yes]

- (1) All assumptions and restrictions are stated clearly and formally. [yes]
- (2) All novel claims are stated formally (e.g., in theorem statements). [yes]
- (3) Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise. [yes]
- (4) Complex formalism, such as definitions or proofs, is motivated and explained clearly. [yes]
- (5) The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision; gratuitous use of mathematical formalism is avoided. [yes]
- (6) Appropriate citations are given for all non-trivial theoretical tools and techniques. [yes]

Articles reporting on computational experiments:

Does this paper include computational experiments? [yes]

- (1) All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. [yes]
- (2) The source code comes with a license that allows free usage for reproducibility purposes. [yes]
- (3) The source code comes with a license that allows free usage for research purposes in general. [yes]
- (4) Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. [yes]
- (5) The unaggregated data comes with a license that allows free usage for reproducibility purposes. [yes]
- (6) The unaggregated data comes with a license that allows free usage for research purposes in general. [yes]
- (7) If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. [yes]
- (8) The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models; amount of memory (cache and RAM); make and version of operating system; names and versions of relevant software libraries and frameworks. [partially]
- (9) The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. [yes]
- (10) The number of algorithm runs used to compute each result is reported. [yes]
- (11) Reported results have not been “cherry-picked” by silently ignoring unsuccessful or unsatisfactory experiments. [yes]
- (12) Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. [yes]
- (13) All (hyper-) parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. [yes]
- (14) The number and range of (hyper-) parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-) parameter optimisation. [NA]
- (15) Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. [NA]

Articles using data sets: Does this work rely on one or more data sets? [no]